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(54) **ELECTRIC FIELD SHAPING LEADS FOR TREATMENT OF CANCER**

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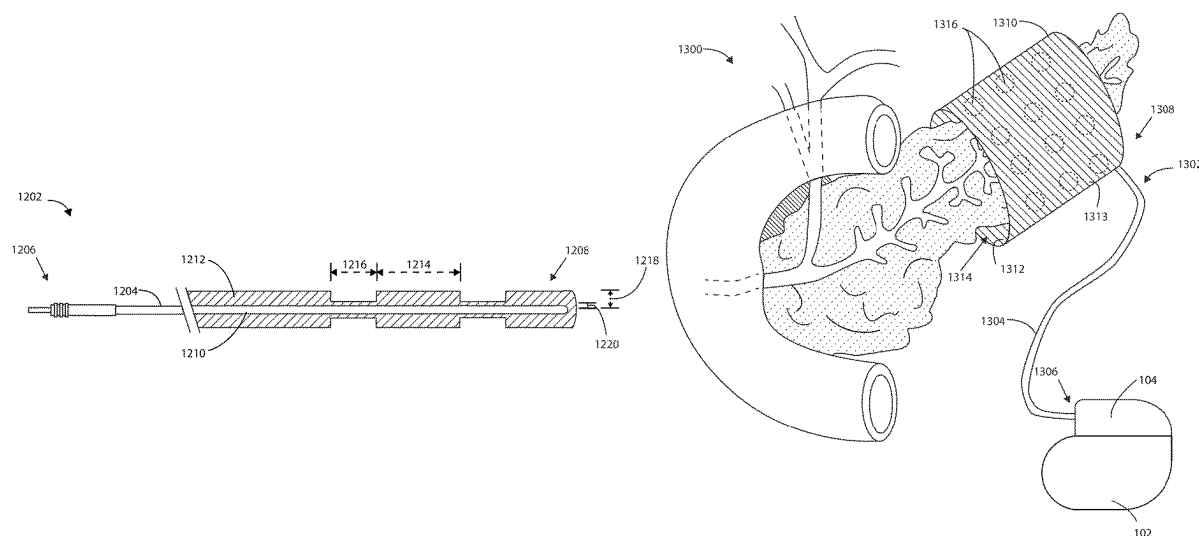
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(57) **ABSTRACT**

Embodiments herein relate to medical devices including electric field shaping leads and methods for using the same to treat cancerous tumors within a bodily tissue. In an embodiment, an implantable lead for a cancer treatment system is disclosed. The lead can include a lead body having a proximal end and a distal end, where the lead body can define a lumen. The lead can also include a paddle disposed at the distal end of the lead body, the paddle having a width that is greater than a width of the lead body. The paddle can include one or more electrodes disposed on the paddle and one or more electrical conductors disposed within the lumen of the lead body to provide electrical communication between the one or more electrodes and the proximal end of the lead body. Other embodiments are also included herein.

19 Claims, 11 Drawing Sheets



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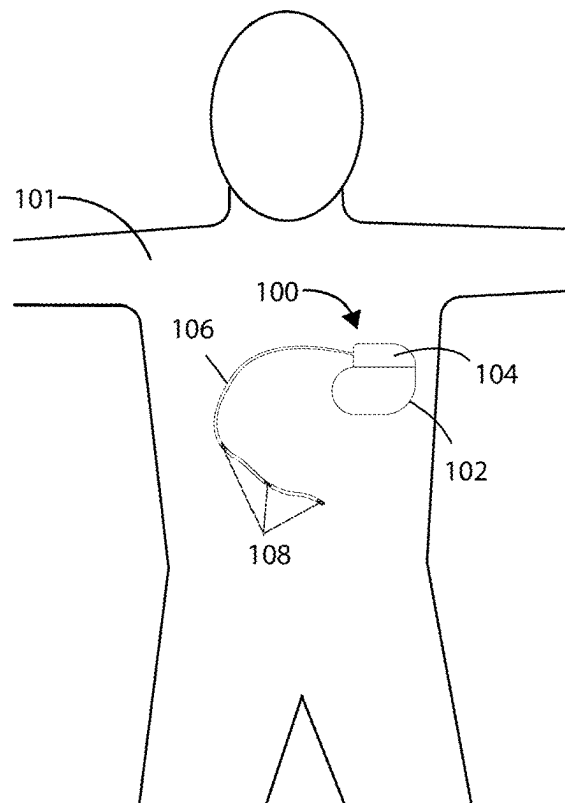


FIG. 1

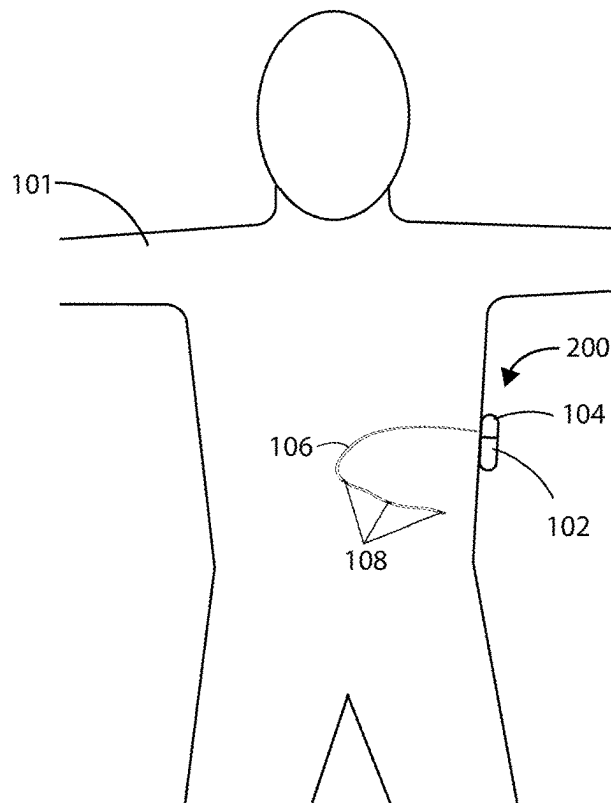


FIG. 2

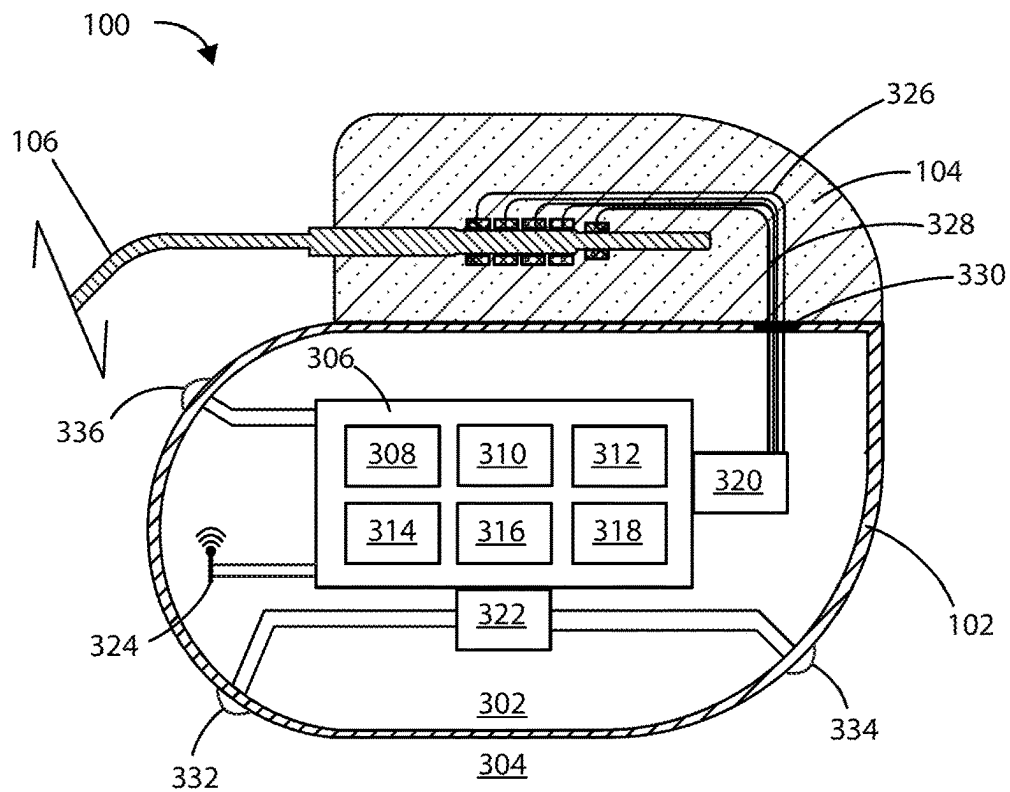


FIG. 3

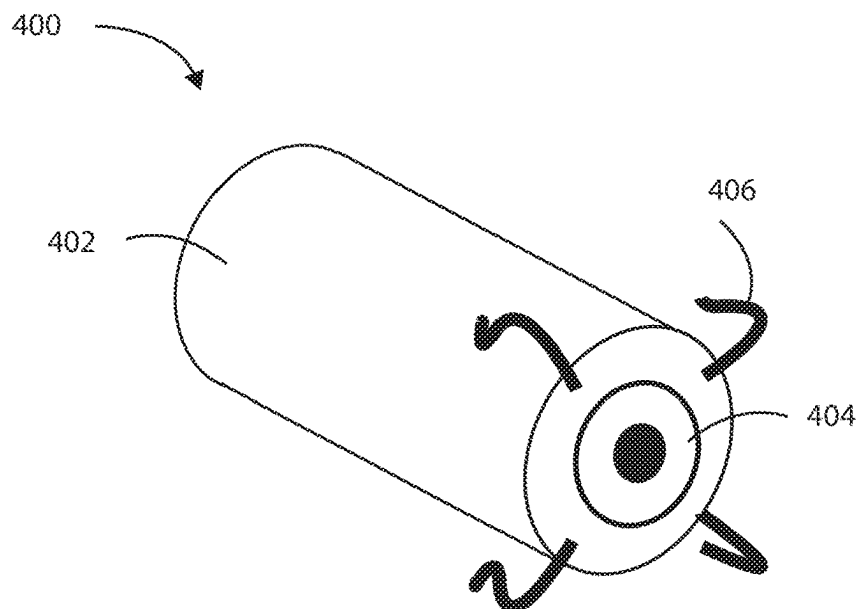


FIG. 4

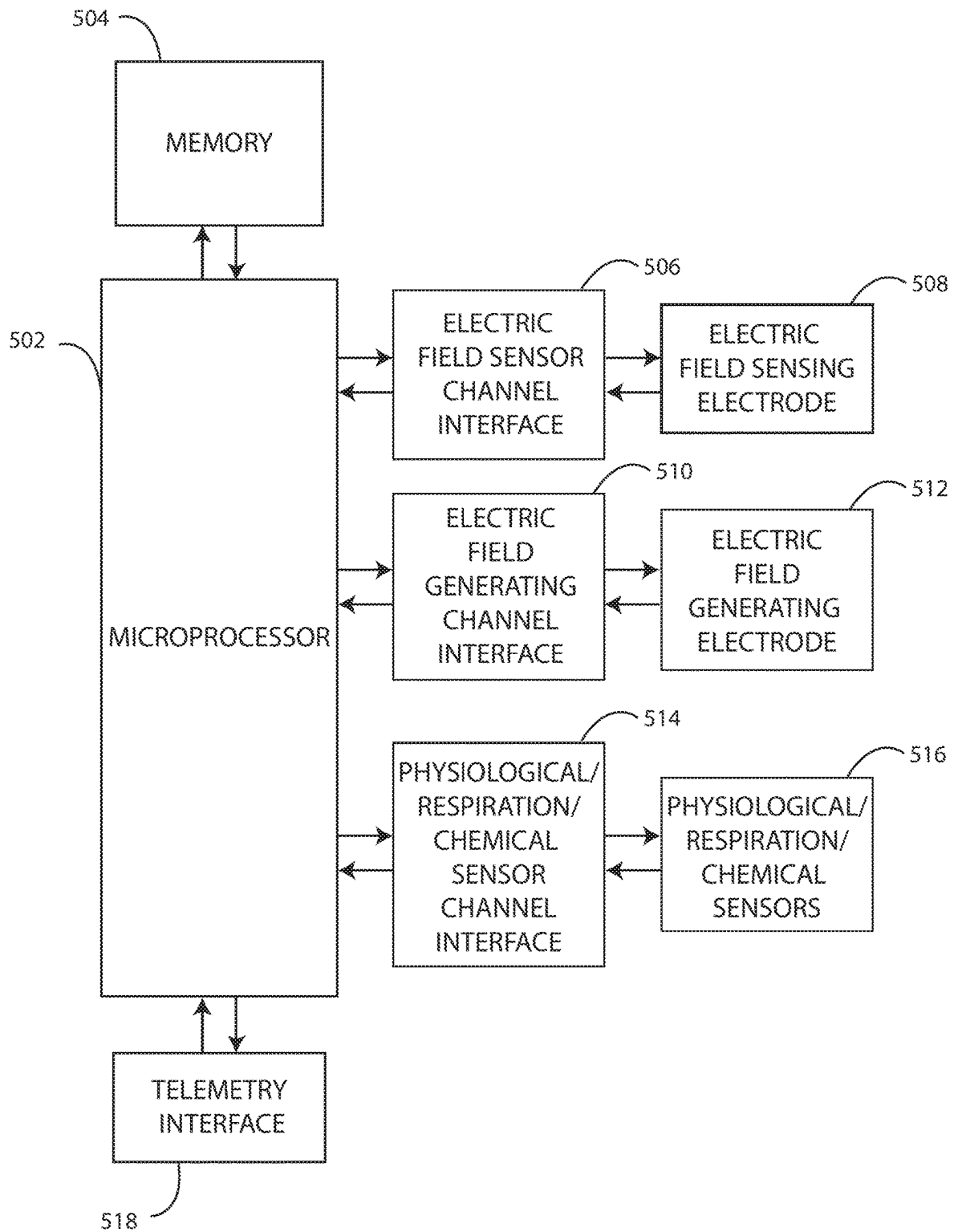


FIG. 5

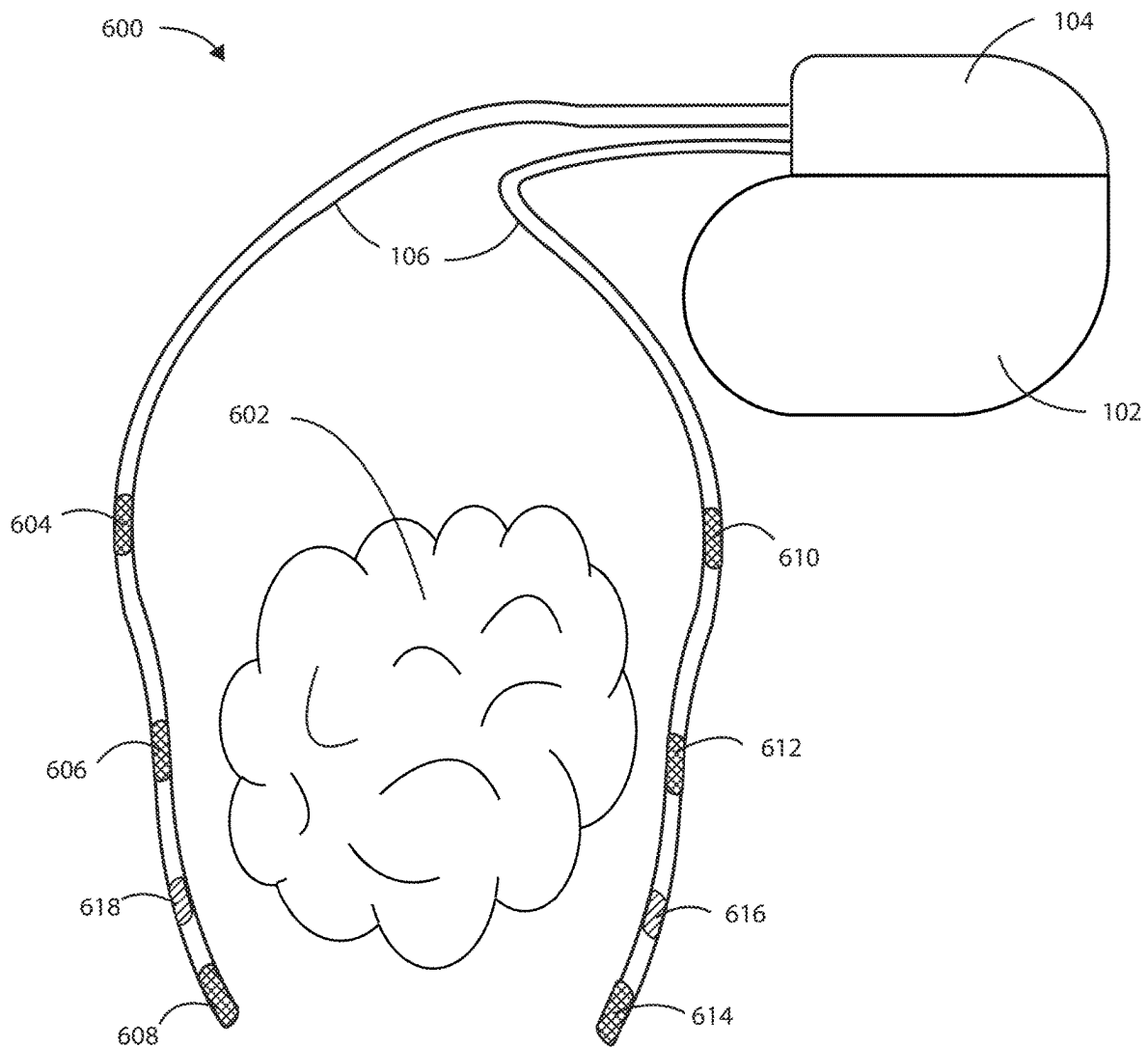


FIG. 6

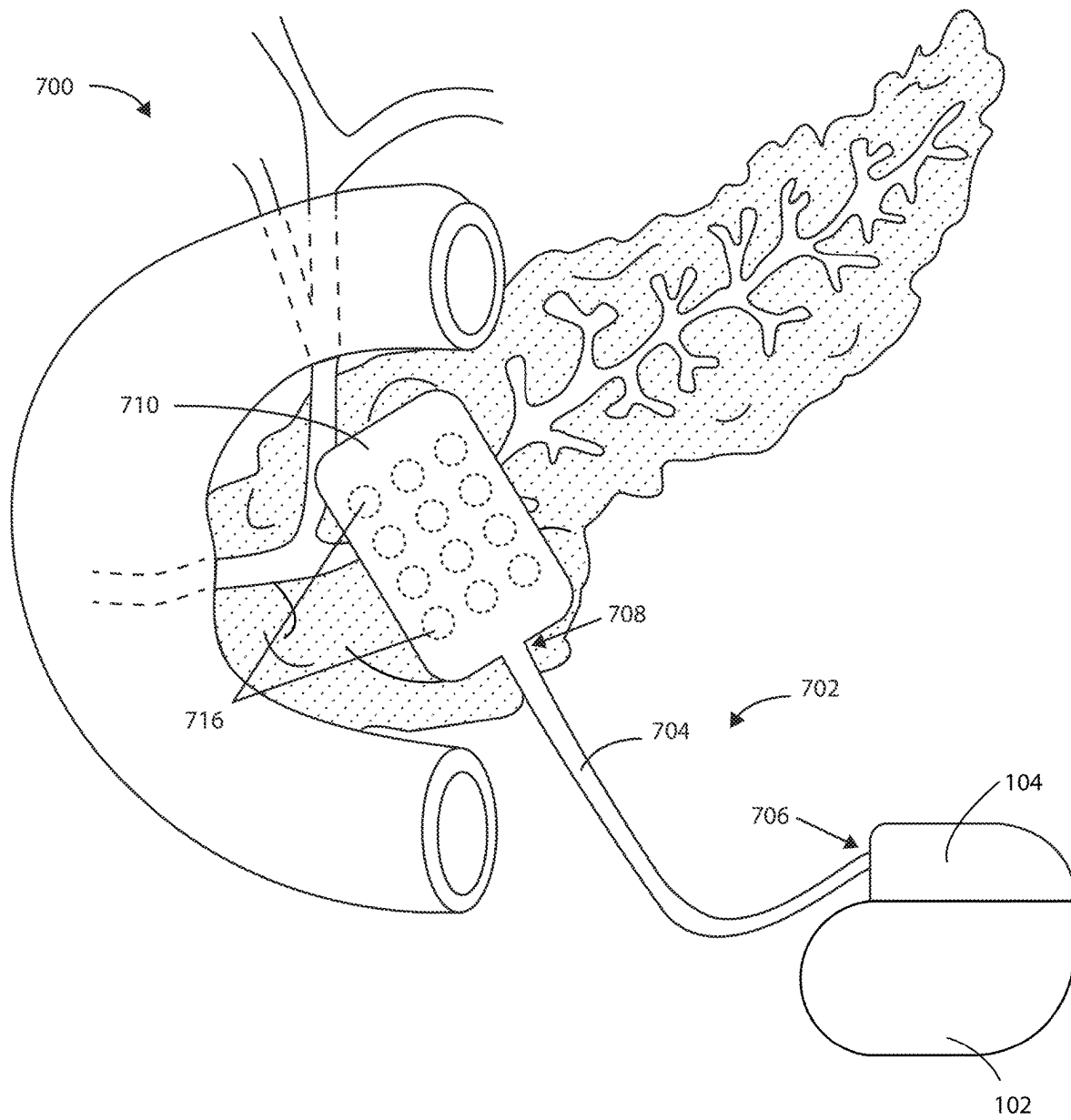


FIG. 7

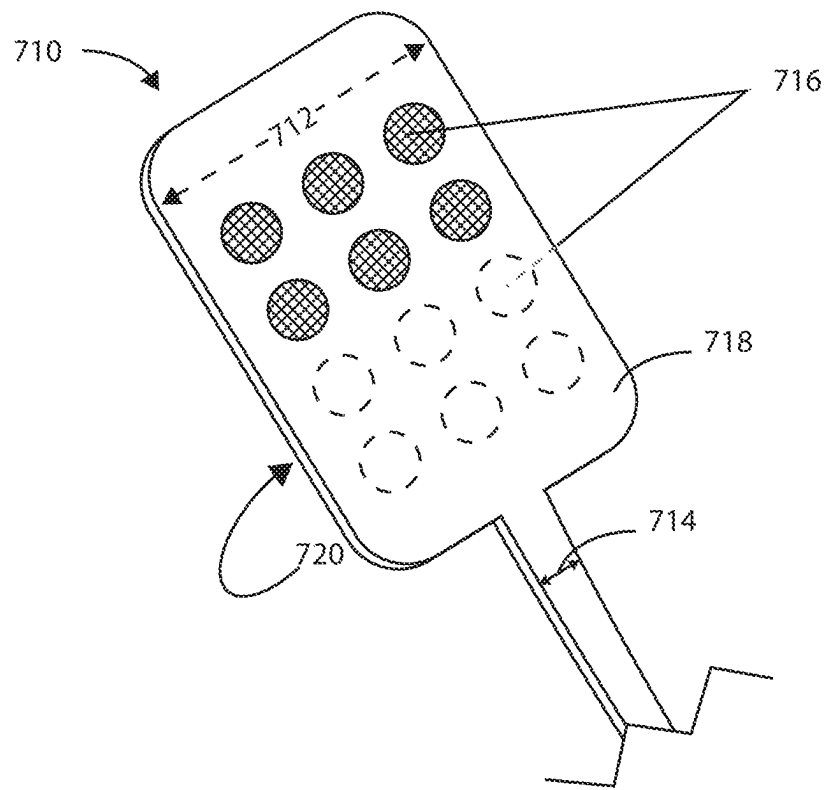


FIG. 8

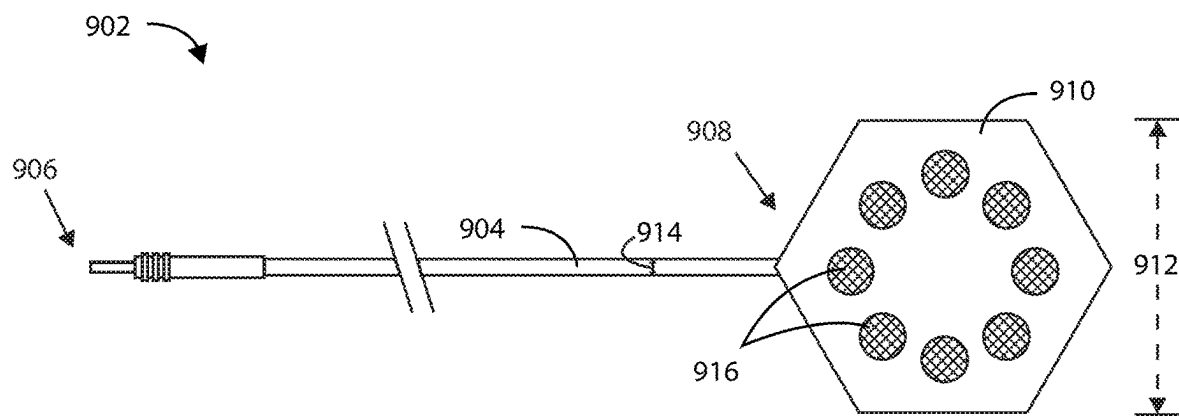


FIG. 9

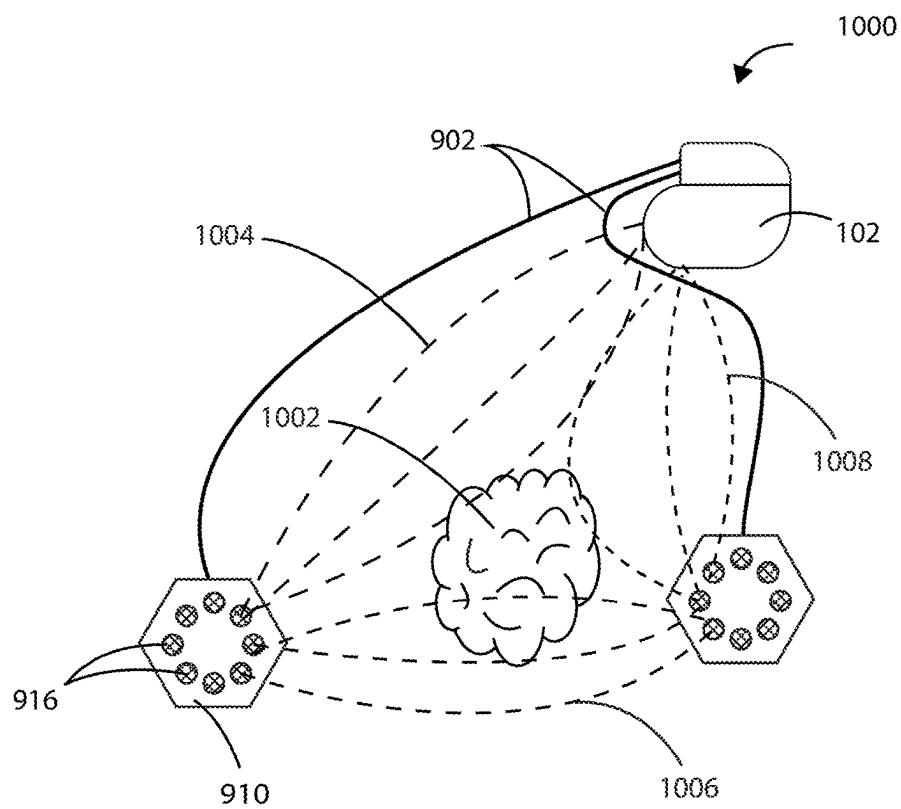


FIG. 10

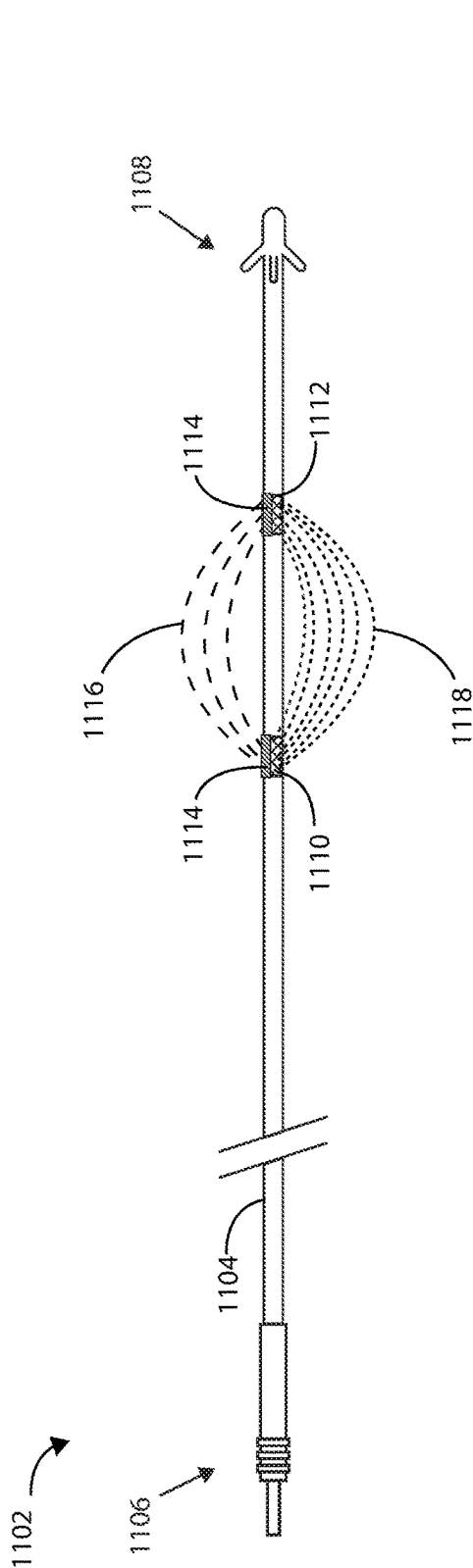


FIG. 11

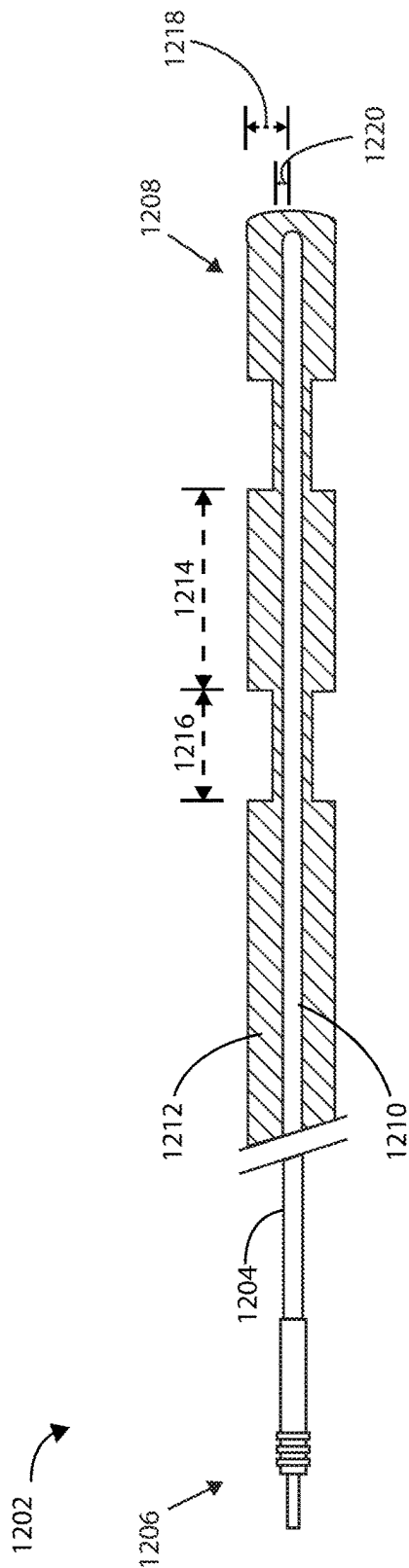


FIG. 12

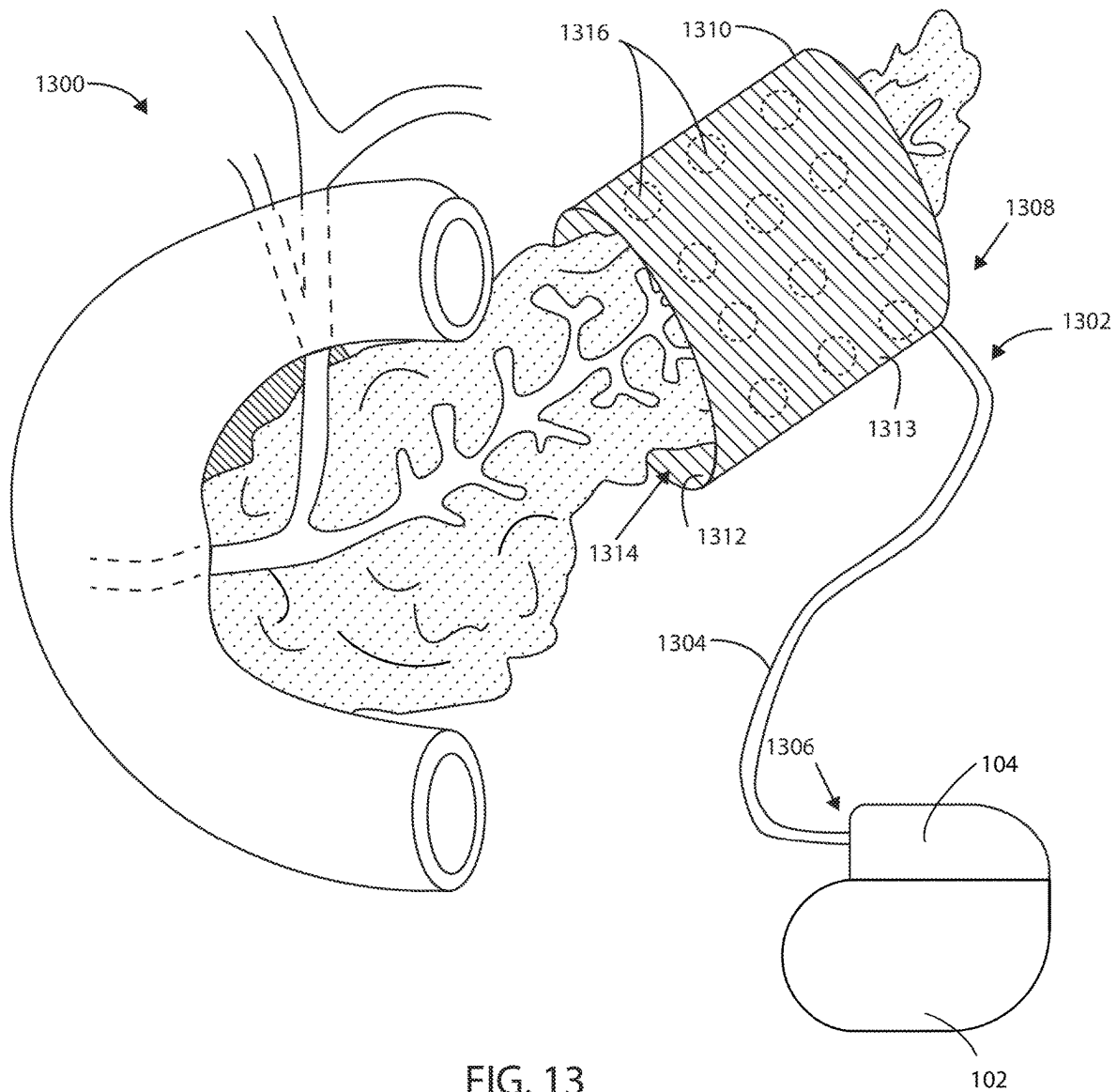


FIG. 13

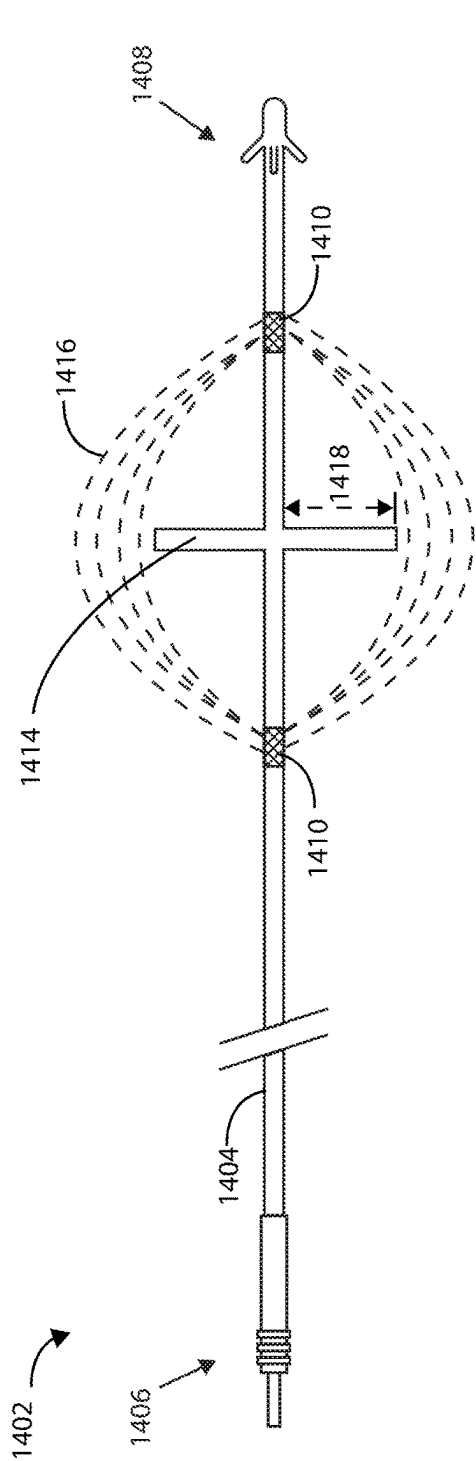


FIG. 14

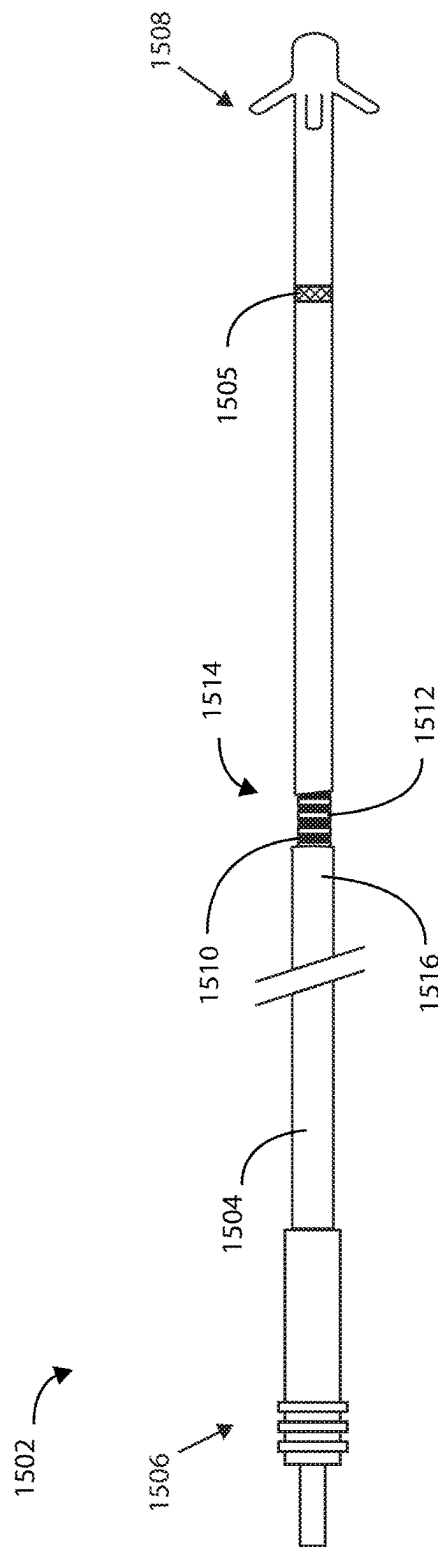


FIG. 15

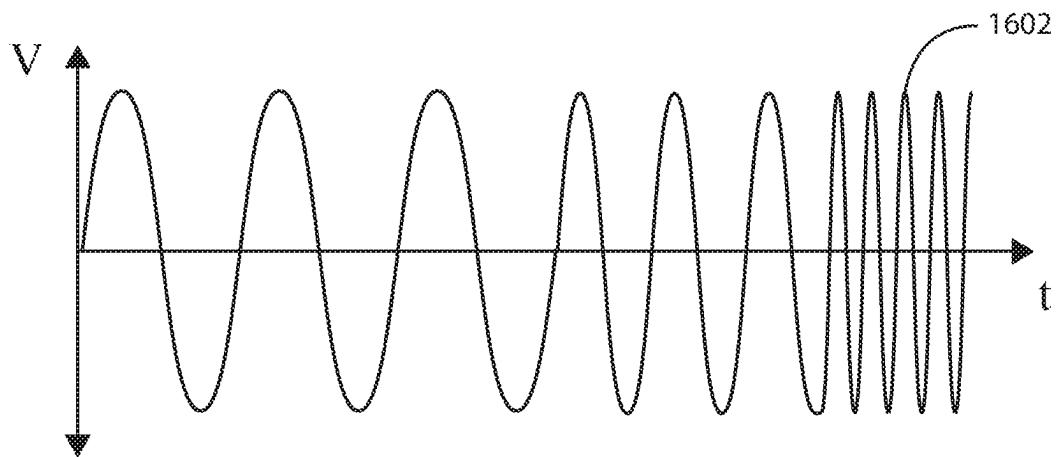


FIG. 16

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ELECTRIC FIELD SHAPING LEADS FOR TREATMENT OF CANCER

This application claims the benefit of U.S. Provisional Application No. 62/575,687, filed Oct. 23, 2017, the content of which is herein incorporated by reference in its entirety.

FIELD

Embodiments herein relate to medical devices including electric field shaping leads and methods for using the same to treat cancerous tumors within a bodily tissue. More specifically, embodiments herein relate to using electric field generating leads configured to generate therapeutic electric fields at the site of a cancerous tumor.

BACKGROUND

According to the American Cancer Society, cancer accounts for nearly 25% of the deaths that occur in the United States each year. The current standard of care for cancerous tumors can include first-line therapies such as surgery, radiation therapy, and chemotherapy. Additional second-line therapies can include radioactive seeding, cryotherapy, hormone or biologics therapy, ablation, and the like. Combinations of first-line therapies and second-line therapies can also be a benefit to patients if one particular therapy on its own is not effective.

Cancerous tumors can form if one normal cell in any part of the body mutates and then begins to grow and multiply too much and too quickly. Cancerous tumors can be a result of a genetic mutation to the cellular DNA or RNA that arises during cell division, an external stimulus such as ionizing or non-ionizing radiation, exposure to a carcinogen, or a result of a hereditary gene mutation. Regardless of the etiology, many cancerous tumors are the result of unchecked rapid cellular division.

Mitosis is the process of cellular division that is a part of the cell cycle for all somatic cells in the body, including many types of cancerous cells. Mitosis includes four basic phases: prophase, metaphase, anaphase, and telophase. Just prior to prophase, a cell will copy its chromosomes to create two identical sister chromatids. During prophase, the chromosomes start to condense and the nuclear membrane surrounding the nucleus disappears. The mitotic spindle also begins to form during prophase. The mitotic spindle includes a self-organized bipolar array of microtubules and centrosomes. Microtubules are generally formed from the polymerization of the highly polar alpha-tubulin and beta-tubulin proteins. Centrosomes are similarly protein-based organelles, two of which migrate to opposite sides of the dividing cell at this phase. The negatively charged end of the microtubules attach to the centrosomes. The positively charged end of the microtubules radiate toward the equator of the dividing cell where they eventually attach to a kinetochore of each sister chromatid. Metaphase can be defined by all chromosomes being aligned at the equator of the dividing cell and bound in the mitotic spindle. An equal number of sister chromatids are then pulled toward opposite ends of the cell during anaphase. Once all chromosomes have been separated, the process of telophase begins, where the cell membrane begins to form a cleavage furrow between the two newly forming sister cells, and cell division becomes complete once the cells physically separate from one another in a process called cytokinesis.

SUMMARY

Embodiments herein include relate to medical devices including electric field shaping leads and methods for using

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the same to treat cancerous tumors within a bodily tissue. In a first aspect, an implantable lead for a cancer treatment system is provided. The implantable lead can include a lead body having a proximal end and a distal end, where the lead body defines a lumen. The implantable lead can also include a paddle disposed at the distal end of the lead body, the paddle having a width that is greater than a width of the lead body. The implantable lead can include one or more electrodes disposed on the paddle and one or more electrical conductors disposed within the lumen of the lead body to provide electrical communication between the one or more electrodes and the proximal end of the lead body.

In a second aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the implantable lead can include from 2 to 36 electrodes disposed on the paddle.

In a third aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the implantable lead can include electrodes arranged in a grid pattern on the paddle.

In a fourth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the implantable lead can be configured to implement a switching mechanism to change which electrodes are currently in electrical communication with the proximal end of the lead body.

In a fifth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the implantable lead can include a paddle having a concave portion.

In a sixth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the implantable lead can include a paddle having a first side configured to face a cancerous tumor and a second side opposite the first side, where one or more electrodes are disposed on the first side.

In a seventh aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a method of treating a cancerous tumor is provided. The method can include implanting a lead within a patient, the lead including a lead body having a proximal end and a distal end and defining a lumen. The lead can include a paddle disposed at the distal end of the lead body, the paddle having a width that is greater than a width of the lead body. The lead can also include one or more electrodes disposed on the paddle and one or more electrical conductors disposed within the lumen of the lead body to provide electrical communication between the one or more electrodes and the proximal end of the lead body. The method can also include generating one or more electric fields at or near the site of a cancerous tumor from the one or more electrodes.

In an eighth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a lead for a cancer treatment system is provided. The lead can include a lead body having a proximal end and a distal end. The lead body can include a patch disposed at the distal end of the lead body, the patch having a width that is greater than the width of the lead body. The patch can be configured to be implanted within a subcutaneous space of a patient. The lead can also include one or more electrodes disposed on the patch and one or more conductors passing through the lead body to provide electrical communication between the one or more electrodes and the proximal end of the lead body.

In a ninth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a method of treating a cancerous tumor is provided.

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The method can include implanting one or more patches subcutaneously within a patient, the one or more patches including one or more electrodes disposed thereon and one or more electrical conductors to provide electrical communication with the one or more electrodes. The method can

also include generating one or more electric fields at or near the site of a cancerous tumor from the one or more electrodes. In a tenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a lead for a cancer treatment system is provided. The lead can include a lead body having a proximal end and a distal end, where the lead body includes one or more electrodes and one or more conductors passing through the lead body to provide electrical communication between the one or more electrodes and the proximal end of the lead body. The lead can also include an insulating material disposed over a portion the one or more electrodes asymmetrically around the diameter of the lead body.

In an eleventh aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a method of treating a cancerous tumor is provided. The method can include implanting a lead within a patient, the lead including a lead body having a proximal end and a distal end. The lead body can include one or more electrodes and one or more conductors passing through the lead body to provide electrical communication between the one or more electrodes and the proximal end of the lead body. The lead can also include an insulating material disposed over a portion the one or more electrodes asymmetrically around the diameter of the lead body. The method can also include generating one or more electric fields at or near the site of a cancerous tumor from the one or more electrodes.

In a twelfth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a lead for a cancer treatment system is provided. The lead can include a lead body having a proximal end and a distal end, the lead body including two or more electrodes and one or more conductors passing through the lead body to provide electrical communication between the two or more electrodes and the proximal end of the lead body. The two or more electrodes can be oriented at different positions radially around the outside surface of the lead body.

In a thirteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a method of treating a cancerous tumor is provided. The method can include implanting a lead within a patient, the lead including a lead body having a proximal end and a distal end. The lead body can include two or more electrodes and one or more conductors passing through the lead body to provide electrical communication between the two or more electrodes and the proximal end of the lead body. The two or more electrodes can be oriented at different positions radially around the outside surface of the lead body. The method can also include generating one or more electric fields at or near the site of to a cancerous tumor from the two or more electrodes.

In a fourteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a lead for a cancer treatment system is provided. The lead can include a lead body having a proximal end and a distal end, the lead body including one or more conductors disposed within the lead body to provide electrical communication. The lead can also include an insulating material disposed over the one or more conductors, the insulating material including one or more first zones and one or more second zones. The thickness of the insulating material over

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the first zones can be greater than the thickness of the insulating material over the second zones. The second zones can function as electrodes to generate one or more electric fields at or near the site of a cancerous tumor.

In a fifteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a method of treating a cancerous tumor is provided. The method can include implanting a lead within a patient, the lead including a lead body having a proximal end and a distal end. The lead body can include one or more conductors disposed within the lead body to provide electrical communication. The lead can also include an insulating material disposed over the one or more conductors, the insulating material including one or more first zones and one or more second zones. The thickness of the insulating material over the first zones can be greater than the thickness of the insulating material over the second zones. The second zones can function as electrodes to generate one or more electric fields at or near the site of a cancerous tumor. The method further including generating one or more electric fields at or near the site of a cancerous tumor from the one or more electrodes.

In a sixteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, an implantable lead for a cancer treatment system is provided. The implantable lead can include a lead body having a proximal end and a distal end. The lead body can define a lumen and can include a cuff disposed at the distal end of the lead body. The cuff can include an inner surface defining an inner cavity and two or more electrodes disposed on the cuff and arrayed on the inner surface of the cuff. The lead can also include one or more electrical conductors disposed within the lumen of the lead body to provide electrical communication between the two or more electrodes and the proximal end of the lead body.

In a seventeenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the implantable lead including from 2 to 36 electrodes disposed on the inner surface of the cuff.

In an eighteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, where the electrodes can be arranged in a grid pattern on the inner surface of the cuff.

In a nineteenth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the implantable lead can be configured to implement a switching mechanism to change which electrodes are currently in electrical communication with the proximal end of the lead body.

In a twentieth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the cuff comprising an elastomeric material.

In a twenty-first aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, where the cuff can expand in diameter by at 50% without structural failure.

In a twenty-second aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the cuff created from a substrate with a plurality of apertures disposed therein.

In a twenty-third aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the cuff created from an expandable woven substrate.

In a twenty-fourth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, an implantable organ cuff for a cancer treatment

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system is provided. The implantable organ cuff can include a cuff body, where the cuff body includes an inner surface that defines an inner cavity. The implantable organ cuff can also include two or more electrodes disposed on the cuff body and arrayed on the inner surface of the cuff body and one or more electrical conductors disposed within or on the cuff body in electrical communication between a connection terminus and the two or more electrodes.

In a twenty-fifth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a method of treating a cancerous tumor is provided. The method can include implanting a lead within a patient, the lead including a lead body having a proximal end and a distal end. The lead body can also include a lumen and can include a cuff disposed at the distal end of the lead body. The cuff can include an inner surface defining an inner cavity. The lead can also include two or more electrodes disposed on the cuff and arrayed on the inner surface of the cuff and one or more electrical conductors disposed within the lumen of the lead body to provide electrical communication between the two or more electrodes and the proximal end of the lead body. The method can also include generating one or more electric fields at or near the site of a cancerous tumor from the one or more electrodes.

In a twenty-sixth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a lead for a cancer treatment system is provided. The lead can include a lead body having a proximal end and a distal end. The lead body can include at least two electrodes and one or more conductors passing through the lead body to provide electrical communication between the at least two electrodes and the proximal end of the lead body. The lead can also include an insulating flange disposed circumferentially on an exterior surface of the lead body, the insulating flange made from an insulating material and configured to inter a direct electrical conduction path between the two electrodes.

In a twenty-seventh aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, the direct electrical conduction path is a straight-line electrical conduction path.

In a twenty-eighth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a lead for a cancer treatment system is provided. The lead can include a lead body having a proximal end and a distal end. The lead body can include a coiled electrode and one or more conductors passing through the lead body to provide electrical communication between the coiled electrode and the proximal end of the lead body. The lead can also include an insulating material disposed over the coiled electrode.

In a twenty-ninth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, where the thickness of the insulating material varies along a length of the coiled electrode.

In a thirtieth aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, where portions of the insulating material are discontinuous creating gaps.

In a thirty-first aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, a method of treating a cancerous tumor is provided. The method can include implanting a lead within a patient, the lead including a lead body having a proximal end and a distal end. The lead body can include a coiled electrode and one or more conductors passing through the lead body to provide electrical communication between the coiled elec-

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trode and the proximal end of the lead body. The lead can also include an insulating material disposed over the coiled electrode. The method can also include generating one or more electric fields at or near the site of a cancerous tumor from the one or more electrodes.

In a thirty-second aspect, in addition to one or more of the preceding or following aspects, or in the alternative to some aspects, where the coiled electrode is implanted subcutaneously.

This summary is an overview of some of the teachings of the present application and is not intended to be an exclusive or exhaustive treatment of the present subject matter. Further details are found in the detailed description and appended claims. Other aspects will be apparent to persons skilled in the art upon reading and understanding the following detailed description and viewing the drawings that form a part thereof, each of which is not to be taken in a limiting sense. The scope herein is defined by the appended claims and their legal equivalents.

BRIEF DESCRIPTION OF THE FIGURES

Aspects may be more completely understood in connection with the following drawings, in which:

FIG. 1 is a schematic view of a medical system in accordance with various embodiments herein.

FIG. 2 is a schematic view of a medical system in accordance with various embodiments herein.

FIG. 3 is a schematic cross-sectional view of a medical device in accordance with various embodiments herein.

FIG. 4 is a schematic view of a medical device in accordance with various embodiments herein.

FIG. 5 is a schematic diagram of components of a medical device in accordance with various embodiments herein.

FIG. 6 is a schematic view of a medical device in accordance with various embodiments herein.

FIG. 7 is a schematic view of a medical device in accordance with various embodiments herein.

FIG. 8 is a schematic view of a portion of a lead in accordance with various embodiments herein.

FIG. 9 is a schematic view of a lead in accordance with various embodiments herein.

FIG. 10 is a schematic view of a medical device in accordance with various embodiments herein.

FIG. 11 is a schematic view of a lead in accordance with various embodiments herein.

FIG. 12 is a schematic view of a lead in accordance with various embodiments herein.

FIG. 13 is a schematic view of a medical device in accordance with various embodiments herein.

FIG. 14 is a schematic view of a lead in accordance with various embodiments herein.

FIG. 15 is a schematic view of a lead in accordance with various embodiments herein.

FIG. 16 is a plot of an exemplary therapy parameter in accordance with various embodiments herein.

While embodiments are susceptible to various modifications and alternative forms, specifics thereof have been shown by way of example and drawings, and will be described in detail. It should be understood, however, that the scope herein is not limited to the particular embodiments described. On the contrary, the intention is to cover modifications, equivalents, and alternatives falling within the spirit and scope herein.

DETAILED DESCRIPTION

As referenced above, many cancerous tumors can result from unchecked rapid cellular division. Some traditional

first-line therapies to treat cancerous tumors can include surgery, radiation therapy, and chemotherapy. However, many first-line therapies have undesirable concomitant side effects, such as fatigue, hair loss, immunosuppression, and long surgical recovery times, to name a few.

While not intending to be bound by theory, it is believed that alternating electric fields can disrupt mitosis within a cancerous tumor by interfering with the dipole alignment of key proteins involved in cellular division; tubulin and septin in particular. The polymerization of tubulin proteins that form microtubule spindle fibers can be disrupted, thus preventing the formation of spindle fibers required for chromosome separation. This can halt cellular division at the metaphase stage of mitosis. In some instances, an alternating electric field can halt polymerization of already growing spindle fibers, leading to incomplete spindles and unequal chromosome separation during anaphase, should the cell survive that long. In each case, halting microtubule spindle formation and unequal chromosome separation during anaphase caused by incomplete polymerization of microtubules can result in apoptosis (i.e., programmed cell death).

It is also believed that alternating electric fields can lead to increased electric field density near the cleavage furrow of the dividing cells during telophase. An increased electric field density in the region of the cleavage furrow can result in dielectrophoresis of charged macromolecules, such as proteins and nucleic acids, toward the high electric field density at the furrow. The unequal concentration of key macromolecules required for cellular division at the site of the cleavage furrow can disrupt the final separation of the sister cells during telophase and eventually lead to apoptosis.

The shape and size of an electric field can be modulated by the positioning of electrodes in space and by varying the electric field at a number of different electrode configurations. Sometimes, the shape of an electric field can be manipulated by alternating or switching the polarity of discrete electrodes within an individual array of electrodes or within the entire medical device system.

Referring now to FIG. 1, a schematic view is shown of a medical device **100** in accordance with various embodiments herein. The medical device **100** can be implanted entirely within the body of a patient **101** at or near the site of a cancerous tumor located within a bodily tissue. Various implant sites can be used including areas such as in the limbs, the upper torso, the abdominal area, the head, and the like.

Referring now to FIG. 2, another schematic view is shown of a medical device **200** in accordance with various embodiments herein. The medical device **200** can be partially implanted within the body of a patient **101**. In some embodiments, the medical device can be partially implanted and partially external to the body of a patient. In other embodiments, a partially implanted medical device can include a transcutaneous connection between components disposed internal to the body and external to the body. A partially implanted medical device can wirelessly communicate with a partially external portion of a medical device over a wireless connection.

In some embodiments, a portion of the medical device can be entirely implanted and a portion of the medical device can be entirely external. For example, in some embodiments, one or more electrodes or leads can be entirely implanted within the body, whereas the portion of the medical device that generates an electric field, such as an electric field generator, can be entirely external to the body. It will be appreciated that in some embodiments described herein, the

electric field generators described can include the many of the same components as and can be configured to perform many of the same functions as a pulse generator. In embodiments where a portion of a medical device is entirely implanted and a portion of the medical device is entirely external, the portion of the medical device that is entirely external can communicate wirelessly with the portion of the medical device that is entirely internal. However, in other embodiments a wired connection can be used.

The medical device **100** or medical device **200** can include a housing **102** and a header **104** coupled to the housing **102**. Various materials can be used. However, in some embodiments, the housing **102** can be formed of a material such as a metal, ceramic, polymer, composite, or the like. In some embodiments, the housing **102**, or one or more portions thereof, can be formed of titanium. The header **104** can be formed of various materials, but in some embodiments the header **104** can be formed of a translucent polymer such as an epoxy material. In some embodiments the header **104** can be hollow. In other embodiments the header **104** can be filled with components and/or structural materials such as epoxy or another material such that it is non-hollow.

In some embodiments where a portion of the medical device **100** or **200** is partially external, the header **104** and housing **102** can be surrounded by a protective casing made of durable polymeric material. In other embodiments, where a portion of the medical device **100** or **200** is partially external, the header **104** and housing **102** can be surrounded by a protective casing made of a combination of polymeric material, metallic material, and/or glass material.

The header **104** can be coupled to one or more leads **106**. The header **104** can serve to provide fixation of the proximal end of one or more leads **106** and electrically couple the one or more leads **106** to one or more components within the housing **102**. The one or more leads **106** can include one or more electrodes **108** disposed along the length of the electrical leads **106**. In some embodiments, electrodes **108** can include electric field generating electrodes and in other embodiments electrodes **108** can include electric field sensing electrodes. In some embodiments, leads **106** can include both electric field generating and electric field sensing electrodes. In other embodiments, leads **106** can include any number of electrodes that are both electric field sensing and electric field generating. It will be appreciated that while many embodiments of medical devices herein are designed to function with leads, leadless medical devices that generate electrical fields are also contemplated herein.

Referring now to FIG. 3, a schematic cross-sectional view of medical device **100** is shown in accordance with various embodiments herein. Housing **102** can define an interior volume **302** that can be hollow and that in some embodiments is hermetically sealed off from the area **304** outside of medical device **100**. In other embodiments the housing **102** can be filled with components and/or structural materials such that it is non-hollow. The medical device **100** can include control circuitry **306**, which can include various components **308**, **310**, **312**, **314**, **316**, and **318** disposed within housing **102**. In some embodiments, these components can be integrated and in other embodiments these components can be separate. In yet other embodiments, there can be a combination of both integrated and separate components. The medical device **100** can also include an antenna **324**, to allow for unidirectional or bidirectional wireless data communication. In some embodiments, the components of medical device **100** can include an inductive energy receiver coil (not shown) communicatively coupled

or attached thereto to facilitate transcutaneous recharging of the medical device via recharging circuitry.

The various components **308**, **310**, **312**, **314**, **316**, and **318** of control circuitry **306** can include, but are not limited to, a microprocessor, memory circuit (such as random access memory (RAM) and/or read only memory (ROM)), recorder circuitry, controller circuit, a telemetry circuit, a power supply circuit (such as a battery), a timing circuit, and an application specific integrated circuit (ASIC), a recharging circuit, amongst others. Control circuitry **306** can be in communication with an electric field generating circuit **320** that can be configured to generate electric current to create one or more fields. The electric field generating circuit **320** can be integrated with the control circuitry **306** or can be a separate component from control circuitry **306**. Control circuitry **306** can be configured to control delivery of electric current from the electric field generating circuit **320**. In some embodiments, the electric field generating circuit **320** can be present in a portion of the medical device that is external to the body.

In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field using one or more frequencies selected from a range of between 10 kHz to 1 MHz. In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field at one or more frequencies selected from a range of between 100 kHz to 500 kHz. In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field at one or more frequencies selected from a range of between 100 kHz to 300 kHz. In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to periodically deliver an electric field using one or more frequencies greater than 1 MHz.

In some embodiments, the electric field can be effective in disrupting cellular mitosis in cancerous cells. The electric field can be delivered to the site of a cancerous tumor along more than one vector. In some examples, the electric field can be delivered along at least one vector, including at least one of the lead electrodes. In some embodiments, at least two vectors with spatial diversity between the two vectors can be used. The vectors can be spatially separated (e.g., the vectors can be disposed at an angle with respect to one another) by at least about 10, 20, 30, 40, 50, 60, 70, 80 or 90 degrees.

A desired electric field strength can be achieved by delivering an electric current between two electrodes. The specific current and voltage at which the electric field is delivered can vary and can be adjusted to achieve the desired electric field strength at the site of the tissue to be treated. In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field using currents ranging from 1 mAmp to 1000 mAmp to the site of a cancerous tumor. In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field using currents ranging from 20 mAmp to 500 mAmp to the site of a cancerous tumor. In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field using currents ranging from 30 mAmp to 300 mAmp to the site of a cancerous tumor.

In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field using currents including 1 mAmp, 2 mAmp, 3 mAmp, 4 mAmp, 5 mAmp, 6 mAmp, 7 mAmp,

8 mAmp, 9 mAmp, 10 mAmp, 15 mAmp, 20 mAmp, 25 mAmp, 30 mAmp, 35 mAmp, 40 mAmp, 45 mAmp, 50 mAmp, 60 mAmp, 70 mAmp, 80 mAmp, 90 mAmp, 100 mAmp, 125 mAmp, 150 mAmp, 175 mAmp, 200 mAmp, 225 mAmp, 250 mAmp, 275 mAmp, 300 mAmp, 325 mAmp, 350 mAmp, 375 mAmp, 400 mAmp, 425 mAmp, 450 mAmp, 475 mAmp, 500 mAmp, 525 mAmp, 550 mAmp, 575 mAmp, 600 mAmp, 625 mAmp, 650 mAmp, 675 mAmp, 700 mAmp, 725 mAmp, 750 mAmp, 775 mAmp, 800 mAmp, 825 mAmp, 850 mAmp, 875 mAmp, 900 mAmp, 925 mAmp, 950 mAmp, 975 mAmp, or 1000 mAmp. It will be appreciated that the control circuitry can be configured to direct the electric field generating circuit **320** to deliver an electric field at a current falling within a range, wherein any of the forgoing currents can serve as the lower or upper bound of the range, provided that the lower bound of the range is a value less than the upper bound of the range.

In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field using voltages ranging from 1 V_{rms} to 50 V_{rms} to the site of a cancerous tumor. In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field using voltages ranging from 5 V_{rms} to 30 V_{rms} to the site of a cancerous tumor. In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field using voltages ranging from 10 V_{rms} to 20 V_{rms} to the site of a cancerous tumor.

In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field using one or more voltages including 1 V_{rms} , 2 V_{rms} , 3 V_{rms} , 4 V_{rms} , 5 V_{rms} , 6 V_{rms} , 7 V_{rms} , 8 V_{rms} , 9 V_{rms} , 10 V_{rms} , 15 V_{rms} , 20 V_{rms} , 25 V_{rms} , 30 V_{rms} , 35 V_{rms} , 40 V_{rms} , 45 V_{rms} , or 50 V_{rms} . It will be appreciated that the control circuitry can be configured to direct the electric field generating circuit **320** to deliver an electric field using a voltage falling within a range, wherein any of the forgoing voltages can serve as the lower or upper bound of the range, provided that the lower bound of the range is a value less than the upper bound of the range.

In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field using one or more frequencies including 10 kHz, 20 kHz, 30 kHz, 40 kHz, 50 kHz, 60 kHz, 70 kHz, 80 kHz, 90 kHz, 100 kHz, 125 kHz, 150 kHz, 175 kHz, 200 kHz, 225 kHz, 250 kHz, 275 kHz, 300 kHz, 325 kHz, 350 kHz, 375 kHz, 400 kHz, 425 kHz, 450 kHz, 475 kHz, 500 kHz, 525 kHz, 550 kHz, 575 kHz, 600 kHz, 625 kHz, 650 kHz, 675 kHz, 700 kHz, 725 kHz, 750 kHz, 775 kHz, 800 kHz, 825 kHz, 850 kHz, 875 kHz, 900 kHz, 925 kHz, 950 kHz, 975 kHz, 1 MHz. It will be appreciated that the electric field generating circuit **320** can deliver an electric field using a frequency falling within a range, wherein any of the foregoing frequencies can serve as the upper or lower bound of the range, provided that the upper bound is greater than the lower bound.

In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to generate one or more applied electric field strengths selected from a range of between 0.25 V/cm to 1000 V/cm. In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to generate one or more applied electric field strengths of greater than 3 V/cm. In some embodiments, the control circuitry **306** can be configured to direct the electric field

generating circuit **320** to generate one or more applied electric field strengths selected from a range of between 1 V/cm to 10 V/cm. In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to generate one or more applied electric field strengths selected from a range of between 3 V/cm to 5 V/cm.

In other embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to generate one or more applied electric field strengths including 0.25 V/cm, 0.5 V/cm, 0.75 V/cm, 1.0 V/cm, 2.0 V/cm, 3.0 V/cm, 5.0 V/cm, 6.0 V/cm, 7.0 V/cm, 8.0 V/cm, 9.0 V/cm, 10.0 V/cm, 20.0 V/cm, 30.0 V/cm, 40.0 V/cm, 50.0 V/cm, 60.0 V/cm, 70.0 V/cm, 80.0 V/cm, 90.0 V/cm, 100.0 V/cm, 125.0 V/cm, 150.0 V/cm, 175.0 V/cm, 200.0 V/cm, 225.0 V/cm, 250.0 V/cm, 275.0 V/cm, 300.0 V/cm, 325.0 V/cm, 350.0 V/cm, 375.0 V/cm, 400.0 V/cm, 425.0 V/cm, 450.0 V/cm, 475.0 V/cm, 500.0 V/cm, 600.0 V/cm, 700.0 V/cm, 800.0 V/cm, 900.0 V/cm, 1000.0 V/cm. It will be appreciated that the electric field generating circuit **320** can generate an electric field having a field strength at a treatment site falling within a range, wherein any of the foregoing field strengths can serve as the upper or lower bound of the range, provided that the upper bound is greater than the lower bound.

In some embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field via leads **106** to the site of a cancerous tumor located within a bodily tissue. In other embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field via the housing **102** of medical device **100** to the site of a cancerous tumor located within a bodily tissue. In other embodiments, the control circuitry **306** can be configured to direct the electric field generating circuit **320** to deliver an electric field between leads **106** and the housing **102** of medical device **100**. In some embodiments, one or more leads **106** can be in electrical communication with the electric field generating circuit **320**. In some embodiments, the one or more leads **106** can include one or more electrodes **108** disposed along the length of the leads **106**, where the electrodes **108** can be in electrical communication with the electric field generating circuit **320**.

In some embodiments, various components within medical device **100** can include an electric field sensing circuit **322** configured to generate a signal corresponding to sensed electric fields. Electric field sensing circuit **322** can be integrated with control circuitry **306** or it can be separate from control circuitry **306**.

Sensing electrodes can be disposed on or adjacent to the housing of the medical device, on one or more leads connected to the housing, on a separate device implanted near or in the tumor, or any combination of these locations. In some embodiments, the electric field sensing circuit **322** can include a first sensing electrode **332** and a second sensing electrode **334**. In other embodiments, the housing **102** itself can serve as a sensing electrode for the electric field sensing circuit **322**. The electrodes **332** and **334** can be in communication with the electric field sensing circuit **322**. The electric field sensing circuit **322** can measure the electrical potential difference (voltage) between the first electrode **332** and the second electrode **334**. In some embodiments, the electric field sensing circuit **322** can measure the electrical potential difference (voltage) between the first electrode **332** or second electrode **334**, and an electrode disposed along the length of one or more leads **106**. In some embodiments, the electric field sensing circuit

can be configured to measure sensed electric fields and to record electric field strength in V/cm.

It will be appreciated that the electric field sensing circuit **322** can additionally measure an electrical potential difference between the first electrode **332** or the second electrode **334** and the housing **102** itself. In other embodiments, the medical device can include a third electrode **336**, which can be an electric field sensing electrode or an electric field generating electrode. In some embodiments, one or more sensing electrodes can be disposed along lead **106** and can serve as additional locations for sensing an electric field. Many combinations can be imagined for measuring electrical potential difference between electrodes disposed along the length of one or more leads **106** and the housing **102** in accordance with the embodiments herein.

In some embodiments, the one or more leads **106** can be in electrical communication with the electric field generating circuit **320**. The one or more leads **106** can include one or more electrodes **108**, as shown in FIGS. **1** and **2**. In some embodiments, various electrical conductors, such as electrical conductors **326** and **328**, can pass from the header **104** through a feed-through structure **330** and into the interior volume **302** of medical device **100**. As such, the electrical conductors **326** and **328** can serve to provide electrical communication between the one or more leads **106** and control circuitry **306** disposed within the interior volume **302** of the housing **102**.

In some embodiments, recorder circuitry can be configured to record the data produced by the electric field sensing circuit **322** and record time stamps regarding the same. In some embodiments, the control circuitry **306** can be hardwired to execute various functions, while in other embodiments the control circuitry **306** can be directed to implement instructions executing on a microprocessor or other external computation device. A telemetry circuit can also be provided for communicating with external computation devices such as a programmer, a home-based unit, and/or a mobile unit (e.g. a cellular phone, personal computer, smart phone, tablet computer, and the like).

Referring now to FIG. **4**, leadless medical device **400** is shown in accordance with the embodiments herein. The leadless medical device **400** can include a housing **402** and a header **404** coupled to the housing **402**. Various materials can be used. However, in some embodiments, the housing **402** can be formed of a material such as a metal, ceramic, polymer, composite, or the like. In some embodiments, the housing **402**, or one or more portions thereof, can be formed of titanium. The header **404** can be formed of various materials, but in some embodiments the header **404** can be formed of a translucent polymer such as an epoxy material. In some embodiments the header **404** can be hollow. In other embodiments the header **404** can be filled with components and/or structural materials such as epoxy or another material such that it is non-hollow. In some embodiments, leadless medical device **400** can include fixation elements **406** to keep a leadless medical device **400** positioned at or near the site of a cancerous tumor within the body. In some embodiments, fixation elements **406** can include talons, tines, helices, bias, and the like.

Elements of various embodiments of the medical devices described herein are shown in FIG. **5**. However, it will be appreciated that some embodiments can include additional elements beyond those shown in FIG. **5**. In addition, some embodiments may lack some elements shown in FIG. **5**. The medical devices as embodied herein can gather information through one or more sensing channels and can output information through one or more field generating channels.

A microprocessor **502** can communicate with a memory **504** via a bidirectional data bus. The memory **504** can include read only memory (ROM) or random access memory (RAM) for program storage and RAM for data storage. The microprocessor **502** can also be connected to a telemetry interface **518** for communicating with external devices such as a programmer, a home-based unit and/or a mobile unit (e.g. a cellular phone, personal computer, smart phone, tablet computer, and the like) or directly to the cloud or another communication network as facilitated by a cellular or other data communication network. In some embodiments, the medical device can include an inductive energy receiver coil interface (not shown) communicatively coupled or attached thereto to facilitate transcutaneous recharging of the medical device.

The medical device can include one or more electric field sensing electrodes **508** and one or more electric field sensor channel interfaces **506** that can communicate with a port of microprocessor **502**. The medical device can also include one or more electric field generating electrodes **512** and one or more electric field generating channel interfaces **510** that can communicate with a port of microprocessor **502**. The medical device can also include one or more physiological sensors, respiration sensors, or chemical sensors **516** and one or more physiological/respiration/chemical sensor channel interfaces **514** that can communicate with a port of microprocessor **502**. The channel interfaces **506**, **510**, and **514** can include various components such as analog-to-digital converters for digitizing signal inputs, sensing amplifiers, registers which can be written to by the control circuitry in order to adjust the gain and threshold values for the sensing amplifiers, source drivers, modulators, demodulators, multiplexers, and the like.

In some embodiments, the physiological sensors can include sensors that monitor temperature, blood flow, blood pressure, and the like. In some embodiments, the respiration sensors can include sensors that monitor respiration rate, respiration peak amplitude, and the like. In some embodiments, the chemical sensors can measure the quantity of an analyte present in a treatment area about the sensor, including but not limited to analytes such as of blood urea nitrogen, creatinine, fibrin, fibrinogen, immunoglobulins, deoxyribonucleic acids, ribonucleic acids, potassium, sodium, chloride, calcium, magnesium, lithium, hydronium, hydrogen phosphate, bicarbonate, and the like. However, many other analytes are also contemplated herein. Exemplary chemical/analyte sensors are disclosed in commonly owned U.S. Pat. No. 7,809,441 to Kane et al., and which is hereby incorporated by reference in its entirety.

Although the physiological, respiration, or chemical sensors **516** are shown as part of a medical device in FIG. **5**, it is realized that in some embodiments one or more of the physiological, respiration, or chemical sensors could be physically separate from the medical device. In various embodiments, one or more of the physiological, respiration, or chemical sensors can be within another implanted medical device communicatively coupled to a medical device via telemetry interface **518**. In yet other embodiments, one or more of the physiological, respiration, or chemical sensors can be external to the body and coupled to a medical device via telemetry interface **518**.

Referring now to FIG. **6**, a schematic diagram of a medical device **600** is shown in accordance with the embodiments herein. Medical device **600** can include housing **102** and header **104**, and one or more leads **106**. Leads **106** can include one or more electrodes such as electrodes **604**, **606**, **608**, **610**, **612**, or **614** disposed along the length of the leads

106. In some embodiments, electrodes **604**, **606**, **608**, **610**, **612**, or **614** can include electric field generating electrodes and in other embodiments electrodes **604**, **606**, **608**, **610**, **612**, or **614** can include electric field sensing electrodes. In some embodiments, leads **106** can include both electric field generating and electric field sensing electrodes.

The proximal ends of leads **106** are disposed within the header **104**. The distal ends of electrical leads **106** can surround a cancerous tumor **602** such that the electrodes **604**, **606**, **608**, **610**, **612**, or **614** are brought into proximity of the cancerous tumor **602**. In some embodiments, the leads **106** can be positioned within the vasculature such that electrodes **604**, **606**, **608**, **610**, **612**, or **614** are adjacent to or positioned within the cancerous tumor **602**. However, it will be appreciated that leads **106** can be disposed in various places within or around the cancerous tumor **602**. In some embodiments, the leads **106** can pass directly through the cancerous tumor **602**.

In some embodiments, the leads **106** can include one or more tracking markers **616** or **618** along the length of the lead for use in determining the precise location of the electrodes relative to the tumor. In some embodiments, the one or more tracking markers can be disposed directly distal or directly proximal to the one or more electrodes disposed on the lead. In some embodiments, the tracking markers can be formed from a magnetic material. In some embodiments, the tracking markers can be formed from a radiographic material. In some embodiments, the tracking markers can be formed from a fluorographic material.

It will be appreciated that a plurality of electric field vectors can be generated between various combinations of electrodes **604**, **606**, **608**, **610**, **612**, or **614** disposed along leads **106** to create an electric field. For example, one or more electric field vectors can be generated between electrodes **604** and **610**. Similarly, one or more electric field vectors can be generated between electrodes **606** and **612**. It will also be appreciated that one or more electric field vectors can be generated between any combination of electrodes **604**, **606**, **608**, **610**, **612**, or **614**. In some embodiments, one or more electric field vectors can be generated between any combination of electrodes **604**, **606**, **608**, **610**, **612**, or **614** and the housing **102** of medical device **400**. It will be appreciated that one or more unipolar or multipolar leads can be used in accordance with the embodiments herein. In some embodiments, a combination of unipolar and multipolar leads can be used. In other embodiments, a circular lead, clamp lead, cuff lead, paddle lead, or patch lead can be used.

Referring now to FIG. **7** (not shown to scale), lead **702** for use in a cancer treatment system **700** is shown in accordance with various embodiments herein. It will be appreciated that cancer treatment system **700** is shown in the context of providing treatment to a site of a cancerous tumor at or near the pancreas, but cancer treatment system **700** can be implanted at any suitable location within the body. Cancer treatment system **700** can include one or more leads **702** for delivering a cancer treatment. The leads **702** can include a lead body **704** having a proximal end **706** and a distal end **708**, where the lead body **704** can define a lumen (not shown). The lead body **704** can include a paddle **710** disposed at the distal end **708** of the lead body **704**. Fixation of the paddle **710** to an organ and/or tumor site can be achieved by a suture, staple, talon, glue, and the like. There can be one or more electrodes **716** disposed on the paddle **710**. There can be one or more electrical conductors (not shown) disposed within the lumen of the lead body **704** to

provide electrical communication between the one or more electrodes **716** and the proximal end **706** of the lead body **704**.

Electric fields can be generated between any combinations of electrodes **716** disposed on paddle **710**. In some embodiments, electric fields can be generated between any combinations of electrodes **716** disposed on two or more paddles **710**. In some embodiments, electric fields can be generated between any combinations of electrodes **716** disposed on one or more paddles **710** and the housing **102** of a medical device. In other embodiments, electric fields can be generated between any combinations of electrodes **716** disposed on one or more paddles **710** and one or more additional leads, such as any of the other lead configurations disclose herein, having electrodes disposed thereon and placed at the site of a cancerous tumor. In some embodiments, the electrodes **716** can be field generating electrodes or field sensing electrodes.

In some embodiments, the one or more leads **702** can include anywhere from 2 to 36 electrodes **716** disposed on the paddle **710**. In some embodiments, the one or more leads **702** can include anywhere from 2 to 50 electrodes **716** disposed on the paddle **710**. In some embodiments, the one or more leads **702** can include anywhere from 3 to 12 electrodes **716** disposed on the paddle **710**. In some embodiments, the one or more leads **702** can include anywhere from 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49 or 50 electrodes **716** disposed on the paddle **710**. It will be appreciated that the one or more leads **702** can include any number of electrodes falling within a range, wherein any of the foregoing numbers of electrodes can serve as the upper or lower bound of the range, provided that the upper bound is greater than the lower bound.

In some embodiments, the shape of the electric field can be influenced by the placement of the electrodes **716** on the paddle **710** and/or by the shape of the paddle **710**. In some embodiments, the electrodes **716** can be arranged in a grid pattern on the paddle **710**. In other embodiments, the electrodes **716** can be arranged in a circular array on the paddle **710**. In some embodiments, the paddle **710** can include a concave portion. In other embodiments, the paddle **710** can include a convex portion.

Referring now to FIG. **8**, the paddle **710** can include a first side **718** configured to face a cancerous tumor and a second side **720** opposite the first side **718**. In some embodiments, the one or more electrodes **716** can be disposed on the first side **718**. In some embodiments, the one or more electrodes **716** can be disposed on the second side **720**, opposite the first side **718**. In other embodiments, the one or more electrodes **716** can be disposed on both the first side **718** and the second side **720**. The paddle **710** can have a width **712** that is greater than a width **714** of the lead body **704**.

To provide a means to modulate the shape and position of the electric field about the site of a cancerous tumor, in some embodiments the one or more leads **702** can be configured to implement a switching mechanism to change which electrodes are currently in electrical communication with the proximal end **706** of the lead body **704**.

The one or more leads **702** can be suitable for use in a method of treating a cancerous tumor, in accordance with the embodiments herein. The method of treating a cancerous tumor can include implanting one or more leads **702** within a patient, where the one or more leads **702** include a lead body **704** having a proximal end **706** and a distal end **708**. The lead body **704** can define a lumen (not shown) and

include a paddle **710** disposed at the distal end **708** of the lead body **704**. As described above, the paddle **710** can have a width **712** that is greater than a width **714** of the lead body **704**. There can be one or more electrodes **716** disposed on the paddle **710**. There can be one or more electrical conductors (not shown) disposed within the lumen of the lead body **704** to provide electrical communication between the one or more electrodes **716** and the proximal end **706** of the lead body **704**. The method of treating a cancerous tumor can also include the step of generating an electric field at the site of a cancerous tumor from the one or more electrodes **706**.

Referring now to FIG. **9**, a lead **902** is shown in accordance with various embodiments herein. The lead **902** can include a lead body **904** having a proximal end **906** and a distal end **908**. The lead body **904** can include a patch **910** disposed at the distal end **908** of the lead body **904**. The patch **910** can have a width **912** that is greater than the width **914** of the lead body **904**. The patch **910** can be configured to be implanted within a subcutaneous space of a patient. Fixation of the patch **910** to an organ and/or tumor site can be achieved by a suture, staple, talon, glue, and the like. There can be one or more electrodes **916** disposed on the patch **910**. There can also be one or more conductors (not shown) passing through the lead body **904** and providing electrical communication between the one or more electrodes **916** and the proximal end **906** of the lead body **904**. In some embodiments, the patch **910** can be implanted subcutaneously.

In some embodiments, the lead **902** can include anywhere from 2 to 36 electrodes **916** disposed on the patch **910**. In some embodiments, the one or more leads **902** can include anywhere from 2 to 50 electrodes **916** disposed on the patch **910**. In some embodiments, the one or more leads **902** can include anywhere from 3 to 12 electrodes **916** disposed on the patch **910**. In some embodiments, the one or more leads **902** can include anywhere from 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49 or 50 electrodes **916** disposed on the patch **910**. In some embodiments, the lead **902** can include one or more electrodes on a first side of patch **910** and one or more electrodes on a second side of patch **910**. It will be appreciated that the one or more leads **902** can include any number of electrodes falling within a range, wherein any of the foregoing numbers can serve as the upper or lower bound of the range, provided that the upper bound is greater than the lower bound.

Referring now to FIG. **10**, one or more leads **902** within cancer treatment system **1000** are shown in accordance with various embodiments herein. Cancer treatment system **1000** can include one or more leads **902** for implementing a cancer treatment for a cancerous tumor **1002**.

Electric fields can be generated between any combinations of electrodes **916** disposed on patches **910** or housing **102**. For example, near-field electric field **1006** can be generated between combinations of electrodes **916** disposed on two or more patches **910**. In some embodiments, far-field electric fields **1004** and **1008** can be generated between any combinations of electrodes **916** disposed on one or more patches **910** and the housing **102** of a medical device. Many electric fields of various electric field strengths and shapes can be generated in cancer treatment system **1000** to provide spatial diversity to the electric fields about cancerous tumor **1002**. It will be appreciated that electrodes used to generate near-field electric fields at or near the site of a cancerous tumor can provide high electric field density at the site of the

tumor. In some embodiments, electrodes on a lead placed transvascularily near the site of a cancerous tumor can provide near-field field density to the cancerous tumor. It will also be appreciated that the electrodes and/or housing used to generate far-field electric fields can increase the spatial diversity of the electric field at the site of the tumor. In some embodiments, one or more subcutaneous patch electrodes can be used to increase the spatial diversity of the electric field at the site of the tumor. In some embodiments, mixtures of near-field and far-field leads can be used.

The one or more leads **902** can be suitable for use in a method for treating a cancerous tumor, in accordance with the embodiments herein. The method of treating a cancerous tumor can include implanting one or more patches **910** subcutaneously within a patient. The one or more patches **910** can include one or more electrodes **916** disposed on the one or more patches. The one or more patches **910** can also include one or more electrical conductors (not shown) providing electrical communication with the one or more electrodes **916**. The method of treating a cancerous tumor can also include generating one or more electric fields at or near the site of a cancerous tumor from the one or more electrodes **916**.

Referring now to FIG. **11**, a lead **1102** for use in a cancer treatment system is shown in accordance with the embodiments herein. The lead **1102** can include a lead body **1104** having a proximal end **1106** and a distal end **1108**. The lead body **1104** can include one or more electrodes, such as electrodes **1110** and **1112**. In some embodiments, the lead body **1104** can include two or more electrodes. The lead **1102** can also include one or more conductors (not shown) passing through the lead body **1104** to provide electrical communication between the one or more electrodes **1110** and **1112** and the proximal end **1106** of the lead body **1104**. In some embodiments, the one or more electrodes can be oriented at different positions radially around the outside surface of the lead body **1104**. In some embodiments, the one or more electrodes can be oriented at the same positions radially around the outside surface of the lead body **1104**.

The electric field strength and direction can be manipulated in or near a tumor by coating all or part of the electrodes with an insulating material **1114**. In some embodiments, the lead **1102** can include an insulating material **1114** disposed over all of one or more electrodes. In some embodiments, the lead **1102** can include an insulating material **1114** disposed over part of one or more electrodes. In some embodiments, the insulating material can be asymmetrically disposed over a portion of the one or more electrodes. In some embodiments, the electrodes can be disposed on the lead **1102** such that the electrodes are pointing in different directions. In other embodiments, the electrodes can be disposed on the lead **1102** such that the electrodes are pointing in the same direction.

The electric field strength can be affected by the placement and thickness of the insulating material **1114**. For example, electric field **1116** shown between electrodes **1110** and **1112** can be weaker than electric field **1118** shown between electrodes **1110** and **1112** due to the insulating material **1114**. Insulating material **1114** can be selected from various materials, including but not limited to electrically insulating polymers such as expanded polytetrafluoroethylene (ePTFE), polytetrafluoroethylene (PTFE), polyethylene-co-tetrafluoroethylene (ETFE), polyurethanes, silicones, poly(p-xylylene) polymers such as parylene polymers, polyether block amides such as PEBAX®, nylons, or derivatives thereof. In some embodiments, the electrodes can be insulated with various materials, including but not limited to

hydrogels or fractal coatings such as iridium oxide, titanium oxide, tantalum pentoxide, other metal oxides, poly(p-xylylene) polymers such as Parylene, and the like.

Electrodes pointing in different directions can generate electric field vectors pointing in different directions, thus enabling the manipulation of the direction of the electric field. In some embodiments, two or more electrodes can be disposed on the same side of the lead **1102**, while in other embodiments two or more electrodes can be disposed on opposite sides of the lead **1102**. In some embodiments the two or more electrodes can be disposed 180 degrees apart. In yet other embodiments the two or more electrodes can be disposed 30 degrees, 45 degrees, 60 degrees, 90 degrees, 120 degrees, or 150 degrees apart as they are disposed radially around the outside surface of the lead body **1104**.

Lead **1102** can be suitable for use in a method for treating a cancerous tumor, in accordance with the embodiments herein. The method of treating a cancerous tumor can include implanting a lead **1102** within a patient. The lead **1102** can include a lead body **1104** having a proximal end **1106** and a distal end **1108**. The lead body **1104** can include one or more electrodes, such as electrodes **1110** and **1112**. In some embodiments, the lead body **1104** can include two or more electrodes. The lead **1102** can also include one or more conductors (not shown) passing through the lead body **1104** to provide electrical communication between the one or more electrodes **1110** and **1112** and the proximal end **1106** of the lead body **1104**. The lead **1102** can also include an insulating material **1114** disposed over a portion the one or more electrodes **1110** and **1112** asymmetrically around the diameter of the lead body. The method of treating a cancerous tumor can also include generating one or more electric fields at or near the site of a cancerous tumor from the two or more electrodes. In some embodiments, the method of treating a cancerous tumor can also include generating an electric field at or near the site of a cancerous tumor from three or more electrodes.

Referring now to FIG. **12**, a lead **1202** for use in a cancer treatment system is shown in accordance with the embodiments herein. The lead **1202** can include a lead body **1202** having a proximal end **1206** and a distal end **1208**. The lead body **1202** can include one or more conductors **1210** disposed within the lead body **1202** to provide electrical communication. The lead body **1202** can also include an insulating material **1212** disposed over the one or more conductors **1210**.

Modifying the thickness of the insulating material **1212** along conductor **1210** so that some regions are thicker and some regions are thinner can provide a mechanism to generate one or more electric fields having one or more electric field strengths in a specific region of interest. In some embodiments, optimal thicknesses for the insulating material **1212** can be less than 10 μm . In some embodiments, optimal thicknesses for the insulating material **1212** can be between about 1 μm to 5 μm . In some embodiments, optimal thicknesses for the insulating material **1212** can be between about 1 μm to 10 μm . In some embodiments, the optimal thicknesses for the insulating material **1212** can be anywhere from 1 μm , 2 μm , 3 μm , 4 μm , 5 μm , 6 μm , 7 μm , 8 μm , 9 μm , and 10 μm . It will be appreciated that the optimal thickness for the insulating material **1212** can fall within a range, wherein any of the foregoing thicknesses can serve as the upper or lower bound of the range, provided that the upper bound is greater than the lower bound.

The insulating material can include one or more first zones **1214** and one or more second zones **1216**. The thickness **1218** of the insulating material **1212** over the first

zones **1214** can be greater than the thickness **1220** of the insulating material **1212** over the second zones **1216**. The second zones **1216** can function as electrodes to generate one or more electric fields at or near the site of a cancerous tumor.

In some embodiments, lead **1202** can be suitable for use in a method for treating a cancerous tumor, in accordance with the embodiments herein. The method of treating a cancerous tumor can include implanting a lead **1202** within a patient. The lead **1202** can include a lead body **1204** having a proximal end **1206** and a distal end **1208**. The lead body **1204** can include one or more conductors **1210** disposed within the lead body **1204** to provide electrical communication. The lead body can also include an insulating material **1212** disposed over the one or more conductors **1210**. The insulating material **1212** can include one or more first zones **1214** and one or more second zones **1216**. The thickness **1218** of the insulating material **1212** over the first zones **1214** is greater than the thickness **1220** of the insulating material **1212** over the second zones **1216**. The second zones can function as electrodes to generate one or more electric fields at or near the site of a cancerous tumor. The method of treating a cancerous tumor can also include generate one or more electric fields at or near the site of a cancerous tumor from the one or more electrodes.

Referring now to FIG. **13**, a cancer treatment system **1300** is shown in accordance with various embodiments herein. It will be appreciated that cancer treatment system **1300** is shown in the context of providing treatment to a site of a cancerous tumor in or near the pancreas, but cancer treatment system **1300** can be implanted at any suitable location within the body. The cancer treatment system **1300** can include a surgically implanted cuff **1310** that can be disposed around an organ containing a solid tumor, or the solid tumor itself.

Cancer treatment system **1300** can include one or more leads **1302** for delivering a cancer treatment. The leads **1302** can include a lead body **1304** having a proximal end **1306** and a distal end **1308**. The lead body **1304** can define a lumen. The lead body **1304** can also include cuff **1310** disposed at the distal end **1308** of the lead body **1304**. Fixation of the cuff **1310** to an organ and/or tumor site can be achieved by a suture, staple, talon, glue, and the like. The cuff **1310** can include an inner surface **1312** defining an inner cavity **1314**. There can be two or more electrodes **1316** disposed on the cuff **1310** and arrayed on the inner surface **1312** of the cuff **1310**. The lead body can also include one or more electrical conductors (not shown) disposed within the lumen of the lead body **1304** to provide electrical communication between the two or more electrodes **1316** and the proximal end **1306** of the lead body **1304**. In some embodiments, one or more electrodes can be disposed on the outer surface **1313**. In some embodiments, one or more electrodes can be disposed on the outer surface **1313** can act as counter electrodes to one or more electrodes disposed on the inner surface **1312**.

In some embodiments, the one or more leads **1302** can include anywhere from 2 to 36 electrodes **1316** disposed on the cuff **1310**. In some embodiments, the one or more leads **1302** can include anywhere from 2 to 50 electrodes **1316** disposed on the cuff **1310**. In some embodiments, the one or more leads **1302** can include anywhere from 3 to 12 electrodes **1316** disposed on the cuff **1310**. In some embodiments, the one or more leads **1302** can include anywhere from 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49 or

50 electrodes **1316** disposed on the cuff **1310**. It will be appreciated that the one or more leads **1302** can include any number of electrodes falling within a range, wherein any of the foregoing numbers of electrodes can serve as the upper or lower bound of the range, provided that the upper bound is greater than the lower bound.

In some embodiments, the electrodes **1316** can be arranged in a grid pattern on the inner surface **1312** of the cuff **1310**. In some embodiments, the electrodes **1316** can be arranged in a circular array on the inner surface **1312** of the cuff **1310**. The electrodes **1316** can be field generating electrodes or field sensing electrodes. The electrodes **1316** can be disposed on the cuff **1310** such that each can be activated independently to generate an electric field of various shapes and sizes. The electrodes **1316** disposed on cuff **1310** can each be activated independently to sense and electric field.

To provide a means to modulate the shape and position of the electric field about the site of a cancerous tumor, in some embodiments the one or more leads **1302** can be configured to implement a switching mechanism to change which electrodes on cuff **1310** are currently in electrical communication with the proximal end **1306** of the lead body **1304**. It will be appreciated that the switching mechanism is controlled by the electric field generating circuit, discussed above with respect to FIG. **3**. Electric field generating circuit can be configured to switch the polarity of one or more electrodes in a multiple electrode array, of three or more electrodes, to change the field shape.

The cuff **1310** can be created from any number of bio-compatible materials. In some embodiments, the cuff **1310** can be made from an elastomeric material. Elastomeric materials suitable for use herein can include, but not be limited to polymers such as expanded polytetrafluoroethylene (ePTFE), polytetrafluoroethylene (ePTFE), polyethylene-co-tetrafluoroethylene (ETFE), polyurethanes, silicones, high-durometer silicones, poly(p-xylylene) polymers such as parylene polymers, polyether block amides such as PEBAX®, nylons, or derivatives thereof. Additionally, the elastomeric materials can be reinforced by materials such as shape-set metals including but not limited to nitinol, to provide additional structural support for clamping down on a tissue containing a cancerous tumor.

It will be appreciated that in some embodiments, the cuff **1310** can expand in diameter by at least 50% without structural failure. It will be appreciated that in other embodiments, the cuff **1310** can expand in diameter by at least 150% without structural failure. In some embodiments, the cuff **1310** can expand in diameter by at least 300% without structural failure. In some embodiments, the cuff **1310** can expand in diameter by 10%, 2%, 30%, 40%, 50%, 60%, 70%, 7%, 80%, 90%, 100%, 12%, 150%, 17%, 200%, 22%, 250%, 27%, 300%, 350%, 400%, 450%, or 500%. It will be appreciated that the cuff **1310** can expand in diameter by any percentage falling within a range of percentages, wherein any of the foregoing percentages can serve as the upper or lower bound of the range, provided that the upper bound is greater than the lower bound.

When implanted, it will be appreciated that cuff **1310** can conform to the shape of the underlying organ or tumor. In some embodiments, the cuff **1310** can include a slit such that it is not a contiguous cylinder. In some embodiments, the cuff **1320** can be reinforced with a shape set material such that it assumes a first elongated rectangular configuration prior to implant, and a second curled cylindrical configuration when implanted around the site of a cancerous tumor.

In some embodiments, the cuff **1310** can include a substrate with a plurality of apertures disposed therein. In some embodiments, the cuff can include an expandable woven substrate. Some non-limiting examples of woven substrate materials can include, but not be limited to aramid, nylons, urethanes, expanded polytetrafluoroethylene (ePTFE), polytetrafluoroethylene (ePTFE), polyethylene-co-tetrafluoroethylene (ETFE), and the like. In some embodiments, the cuff **1310** can be a contiguous cylinder. Some additional examples of non-woven substrate materials can include those created with electrospin processes using polymers such as aramid, nylons, urethanes expanded polytetrafluoroethylene (ePTFE), polytetrafluoroethylene (ePTFE), polyethylene-co-tetrafluoroethylene (ETFE), and the like.

Lead **1302** can be suitable for use in a method for treating a cancerous tumor, in accordance with the embodiments herein. The method of treating a cancerous tumor can include implanting a lead **1302** within a patient and fixing the cuff **1310** at the site of a cancerous tumor. The lead **1302** can include a lead body **1304** having a proximal end **1306** and a distal end **1308**. The lead body can include a lumen and a cuff disposed at the distal end **1308** of the lead body **1304**. The cuff **1310** can include an inner surface **1312** defining an inner cavity **1314**. There can be two or more electrodes **1316** disposed on the cuff **1310** and arrayed on the inner surface **1312** of the cuff **1310**. The lead body **1304** can also include one or more electrical conductors (not shown) disposed within the lumen of the lead body **1304** to provide electrical communication between the two or more electrodes **1316** and the proximal end **1308** of the lead body **1304**. The method of treating a cancerous tumor can also include generating one or more electric fields at or near the site of a cancerous tumor from the one or more electrodes.

Referring now to FIG. **14**, a lead **1402** for use in a cancer treatment system is shown in accordance with the embodiments herein. The lead **1402** can include a lead body **1404** having a proximal end **1406** and a distal end **1408**. The lead body **1404** can include at least two electrodes **1410**. The lead body **1404** can also include one or more conductors (not shown) passing through the lead body **1404** to provide electrical communication between the at least two electrodes **1410** and the proximal end **1408** of the lead body **1404**. The lead body **1404** can also include an insulating flange **1414** disposed circumferentially on an exterior surface of the lead body, the insulating flange **1414** can be made from an insulating material that can interrupt an ionic conduction path **1416** between the two electrodes **1410**. In some embodiments without an insulating flange **1414**, the direct electrical conduction path **1416** can be a straight-line ionic conduction path. In some embodiments with an insulating flange **1414**, the direct ionic conduction path **1416** is a curved path.

In some embodiments, the insulating flange **1414** can be configured about lead **1402** to direct a high field density farther away from the surface of the lead than what would otherwise be found in a straight line path between two or more electrodes. The insulating flange **1414** can be selected from various materials, including but not limited to electrically insulating polymers such as expanded polytetrafluoroethylene (ePTFE), polyurethanes, silicones, and the like. In some embodiments, insulating flange **1414** can be flexible so as to facilitate ease of delivery through the vascular system. In some embodiments, upon delivery of the lead **1402** within the vascular system near the site of a cancerous tumor, insulating flange **1414** can expand within a vessel such that it acts to occlude the passage of blood to prevent low impedance electrical communication through the blood.

Insulating flange **1414** can extend about the circumference of lead **1402** by radial distance **1418**. In some embodiments, radial distance **1418** can be identical all the way around the circumference of lead **1402**. In other embodiments, radial distance **1418** can be a first radial distance about half of the circumference and a second radial distance about the opposite half of the circumference about the lead **1402**. The radial distance **1418** can extend from the surface of lead **1402** by anywhere from 1 mm to 50 mm in length. It will be appreciated that the radial distance **1418** can be selected from a range of distances including 1 mm, 2 mm, 3 mm, 4 mm, 5 mm, 6 mm, 7 mm, 8 mm, 9 mm, 10 mm, 11 mm, 12 mm, 13 mm, 14 mm, 15 mm, 16 mm, 17 mm, 18 mm, 19 mm, 20 mm, 21 mm, 22 mm, 23 mm, 24 mm, 25 mm, 26 mm, 27 mm, 28 mm, 29 mm, 30 mm, 31 mm, 32 mm, 33 mm, 34 mm, 35 mm, 36 mm, 37 mm, 38 mm, 39 mm, 40 mm, 41 mm, 42 mm, 43 mm, 44 mm, 45 mm, 46 mm, 47 mm, 48 mm, 49 mm, 50 mm.

Referring now to FIG. **15**, a lead **1502** is shown in accordance with the embodiments herein. The lead **1502** can include a lead body **1504** having a proximal end **1506** and a distal end **1508**. The lead body **1504** can include one or more electrodes, such as ring electrode **1505**. The lead **1502** can include a conductor core that can be created from one or more sets of wound coils having one, two, three, four, five, or more individual coils wound about a common axis. The coil shown in FIG. **14** is a bifilar coil that includes two individual coils **1510** and **1512** wound about one another. In some embodiments the coil can be a trifilar coil. Insulating material **1516** is shown in a cut-away view, but it will be appreciated that the insulating material **1516** can cover the entire length of the lead **1502**. In some embodiments, the insulating material **1516** has been cut away at locations **1514**. In some embodiments, locations **1514** can serve as an electrode.

In some embodiments, it is useful to pair an electrode, such as an electrode **1514**, distally with a counter electrode at or near the tumor in order to provide spatial diversity of the electric field. Multiple electrodes and counter electrodes placed near a tumor can provide high electric field density at the site of the tumor through use of near-field electrodes. Using the housing of the medical device and/or an electrode placed distal to the tumor can increase the spatial diversity of the electric field at the site of the tumor through use of far-field electrodes.

Leads and Electrodes

The leads described herein can be placed into the body near the site of a cancerous tumor using a number of techniques. Placement of one or more leads can include using techniques such as transvascular placement, tunneling into the subcutaneous space, and/or surgical placement. In some embodiments, the placement of one or more leads can include placement via one or more natural body orifices. The leads can be placed adjacent to or within a cancerous tumor. In some embodiments, multiple leads can be used near to or far from the cancerous tumor.

In some embodiments one or more leads described herein can be placed in the subcutaneous space. Electrodes on leads placed in the subcutaneous space can be used as the primary near-field generating electrode or as a far-field field generating electrode. In some embodiments, electrodes on leads placed in the subcutaneous space can be used as the primary near-field generating electrode or as a far-field field generating electrode in conjunction with the housing of a medical device. Likewise, one or more leads can be placed transvascularly to act as far-field field generating electrodes in

conjunction with an electrode at or near the site of the cancerous tumor or in conjunction with the housing of a medical device.

The leads and electrodes described herein can include additional functional and structural features. In some embodiments, the leads can include those that are compatible with imaging and treatment techniques, including but not limited to MRI (magnetic resonance imaging), X-ray imaging, deep brain stimulation techniques, and/or radiation therapy. In some embodiments, the leads can include one or more conductor cores made from conducting materials. The conductor cores can be formed from conducting materials including metals and/or other conducting materials. Metals can include, but are not limited to, palladium, platinum, silver, gold, copper, aluminum, various alloys including stainless steel, nickel-cobalt alloys such as MP35N® and the like. In some embodiments, the conductor core can be a multifilar coil, including but not limited to a bifilar coil, a trifilar coil, and a quadfilar coil.

In some embodiments, electrodes can be disposed along the length of one or more leads as described herein. Suitable materials for use in the electrodes described herein can include metals such as palladium, to minimize coupling and artifact generation in magnetic fields. In some embodiments, electrodes can be made from other metals and/or other conducting materials. Metals can include, but are not limited to, palladium, platinum, platinum alloys such as platinum-iridium alloy, gold, copper, tantalum, titanium, various alloys including stainless steel, and the like. In some embodiments, electrodes can be in the form of wound coils that can provide an added benefit of increased surface area without compromising flexibility of the electrodes. In some embodiments, the implantable device housing can serve as an electrode.

The leads described herein can also include one or more electrodes disposed along the length of the lead. The leads can include two or more electrodes disposed along the length of the lead. In some embodiments, the electrodes can be tip electrodes found at the distal end of the lead. In other embodiments, the electrodes can be ring electrodes found along the lead but not at the tip of the lead. In some embodiments, the electrodes can be coil electrodes. In some embodiments, a ring or tip electrode can be positioned in or adjacent to a tumor or cancerous tissue and a coil electrode can be positioned farther from the tumor or cancerous tissue in order to help provide spatial diversity to the generated electric fields. In some embodiments, one or more electrodes can have a length along the lengthwise axis (e.g., proximal to distal axis) of about 0.5, 1, 1.5, 2, 3, 4, 5, 7.5, 10, 15, 20, 30, 40, 50, 75, 100 mm or more. In some embodiments, one or more of the electrodes can have a length falling within a range wherein any of the foregoing distances can serve as the upper or lower bound of the range, provided that the upper bound is greater than the lower bound.

The leads can be unipolar, bipolar, or multipolar. In some embodiments, a unipolar lead can include a lead that generates an electric field between one electrode and the housing of the medical device. In some embodiments, a bipolar lead can include a lead that can generate an electric field between two electrodes disposed along the lead, or between both electrodes and the housing of the medical device. In some embodiments, a multipolar lead can include a lead that can generate an electric field between the more than two electrodes disposed along the lead, between more than two electrodes and the housing of the medical device, or any number of combinations of configurations of electrodes and the housing of the medical device.

The electrodes suitable for use here can be made of conductive polymers such as carbon filled silicone, polyacetylene, polypyrrole, polyaniline, polythiophene, polyfuran, polyisoprene, polybutadiene, polyparaphenylene, and the like. In other embodiments, the electrodes can be insulated. In some embodiments, the insulation surrounding and electrode can include microporous insulators to prevent cellular apposition, yet still allow for current flow. Microporous insulators can be made from a number of the insulating materials described herein, including but not limited to polytetrafluoroethylene (ePTFE), polyethylene-co-tetrafluoroethylene (ETFE), polyurethanes, silicones, poly(p-xylylene) polymers such as Parylene polymers, polyether block amides such as PEBAX®, nylons, or derivatives thereof. In some embodiments, the electrodes can be coated with various materials, including but not limited to hydrogels or fractal coatings such as iridium oxide, titanium oxide, tantalum pentoxide, other metal oxides, poly(p-xylylene) polymers such as Parylene, and the like.

A number of lead fixation techniques and configurations can be used in accordance with the embodiments herein. Some non-limiting examples of lead fixation techniques can include biocompatible glue fixation, talon fixation, helix coil fixation, passive centering of the lead in the vascular system, fine fixation within the localized vascular system, spiral bias fixation within the localized vascular system, compression fixation, suture sleeve fixation, and the like. In some examples, the leads embodied herein can be placed within the vascular system surrounding or adjacent to the site of the cancerous tumor. In other embodiments, the leads embodied herein can be placed surgically at or within or surrounding the site of the cancerous tumor.

The leads suitable for use herein can also include one or more open lumens that run the entire longitudinal length of, or a select portion of the longitudinal length of the lead. In some embodiments, the open lumen can include an integrated biopsy apparatus suitable for obtaining biopsy samples from a cancerous tumor site on a periodic basis to monitor disease progression and/or regression. Leads having an open lumen can also be configured to include an integrated drug delivery lumen that can deliver one or more drugs, such as steroids or chemotherapy agents, to the site of the tumor in a single bolus or periodically via a metered pump. The leads can include one or more portals disposed along the length of the lead to provide an outlet for drug delivery at or near the site of a cancerous tumor.

In some embodiments a portion of the lead or the entire lead can include a drug eluting coating. In some embodiments, the drug eluting coating can include an anti-inflammatory agent, such as a steroid. In some embodiments, the steroid can be dexamethasone. In other embodiments, the drug eluting coating can include a chemotherapy agent. In some embodiments, the chemotherapy agent can include a taxane or derivatives thereof, including but not limited to paclitaxel, docetaxel, and the like. In other embodiments, the drug eluting coating can be configured to release additional classes of chemotherapy agents, including, but not limited to alkylating agents, plant alkaloids such as vinca alkaloids, cytotoxic antibiotics, topoisomerase inhibitors, and the like. In some embodiments, the drug eluting coating can be configured to release the drug from the coating in a time-release fashion.

The leads herein can adopt a number of shapes or configurations. In some embodiments, the leads can be linear and in other embodiments the leads can be circular. A circular lead may be a completely closed loop or it may be a semi-closed loop. In some embodiments, the lead can

include a bendable core that can allow the lead to be shaped into many configurations, including but not limited to a U shape, an S shape, a spiral shape, a half circle, an oval, and the like.

In yet other examples, the leads suitable for use herein can include fluorimetric or magnetic markers that can assist the clinician in precise placement at or near the site of a cancerous tumor. The leads can also include integrated pH sensors for detecting the change in the pH at or near the cancerous tumor or other chemical sensors suitable for analyzing the concentration of a chemical analyte of interest.

Therapy Parameters

Successful treatment of cancerous tumors can depend on a number of variables, including electric field strength, frequency, cell heterogeneity, cell size, cancer cell type, tumor size, and location within the body. A variety of therapy parameters can be implemented using the medical devices described herein. One or more therapeutic parameter sets can be programmed into the memory of the medical devices and implemented by the control circuitry **306**, shown in FIG. 3. Exemplary therapeutic parameter sets can include those that implement the following concepts: sweeping through a range of frequencies; stacking of one or more frequencies simultaneously; stepping through one or more frequencies sequentially; the spatial or temporal delivery of one or more electric fields; sweeping through a range of electric field strengths; applying an effective spinning electric field; modulating a voltage control mode or a current control mode; implementing one or more duty cycles; pulse width modulation; manipulation of the waveform shape and/or pulse sequence; and the occasional use of high frequency or high electric fields strength pulses.

The therapeutic parameter sets can be programmed into a medical device to operate autonomously, or they can be queried and manipulated by the patient or a clinician using an external computation device such as a programmer, a home-based unit, and/or a mobile unit (e.g. a cellular phone, personal computer, smart phone, tablet computer, and the like). In other embodiments, the therapeutic parameter sets can be wirelessly communicated to the medical device from an external computation device. Frequencies and/or electric field strengths suitable for use in any of the therapeutic parameter sets herein are discussed above with respect to electric field generating circuit **320**. In some embodiments, one or more therapeutic parameter sets can be implemented simultaneously. In other embodiments, one or more therapeutic parameter sets can be implemented in an alternating fashion.

Referring now to FIG. 16, exemplary plot **1602** shows an example of sweeping through a range of frequencies at the site of a cancerous tumor. Plot **1602** shows an alternating electric field, where the frequency is increased over time as the therapy is applied to the cancerous tumor. In some embodiments, a frequency sweep can include alternating between a first frequency sweep covering a range of about 100 kHz to 300 kHz and a second frequency sweep covering a range about 200 kHz to 500 kHz. It will be appreciated that sweeping through a first and second frequency range as described can be performed indefinitely throughout the course of the therapy.

Electric Field Generators

The medical devices embodied herein can include electric field generators particularly suited for therapeutic and diagnostic techniques used during the course of treatment for a cancerous tumor. In some embodiments, the electric field generators suitable for use herein can include those that have been treated by radiation hardening to make the components

resistant to the damaging effects of radiation therapy treatments often prescribed as a main line treatment for cancerous tumors. Electric field generators can include components such as those described in reference to FIGS. 3 and 5 above.

Electric field generators embodied herein can be programmed with any number of therapeutic parameter sets as described. The electric field generators can be programmed prior to implant, or they can be programmed by a clinician using an external computation device such as a programmer, a home-based unit, and/or a mobile unit (e.g. a cellular phone, personal computer, smart phone, tablet computer, and the like). In some embodiments, therapy parameters can be delivered to the electric field generator via a telemetry circuit. In some embodiments, the electric field generator can include a recharge circuit communicatively coupled to a receiver coil to facilitate transcutaneous recharging of the medical device. In some embodiments, the electric field generator can communicate wirelessly between the receiver coil and an external charging device.

It should be noted that, as used in this specification and the appended claims, the singular forms “a,” “an,” and “the” include plural referents unless the content clearly dictates otherwise. Thus, for example, reference to a composition containing “a compound” includes a mixture of two or more compounds. It should also be noted that the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

It should also be noted that, as used in this specification and the appended claims, the phrase “configured” describes a system, apparatus, or other structure that is constructed or configured to perform a particular task or adopt a particular configuration to. The phrase “configured” can be used interchangeably with other similar phrases such as arranged and configured, constructed and arranged, constructed, manufactured and arranged, and the like.

All publications and patent applications in this specification are indicative of the level of ordinary skill in the art to which this invention pertains. All publications and patent applications are herein incorporated by reference to the same extent as if each individual publication or patent application was specifically and individually indicated by reference.

Aspects have been described with reference to various specific and preferred embodiments and techniques. However, it should be understood that many variations and modifications may be made while remaining within the spirit and scope herein.

The invention claimed is:

1. A lead for a cancer treatment system comprising:

a lead body having a proximal end and a distal end, the lead body having a lumen with a cuff disposed at the distal end of the lead body, the lead body comprising one or more conductors disposed within the lead body and providing electrical communication; and
an insulating material disposed circumferentially about and along the lead body over the one or more conductors;

wherein the lead body comprises one or more first zones and one or more second zones, wherein the insulating material disposed over the one or more conductors in the first zones is thicker than the insulating material disposed over the one or more conductors in the second zones;

wherein the second zones comprise at least a portion of the one or more conductors and at least a portion of the insulating material to form an electric field generating

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electrode configured to generate one or more therapeutic electric fields at or near a site of a cancerous tumor; wherein the portion of the insulating material disposed about the one or more conductors in the second zones is configured to manipulate a strength of the one or more therapeutic electric fields generated by the electric field generating electrodes within the second zones; the second zones having a stronger electric field than the first zones; and

wherein the cuff is reinforced with a shape set material such that the cuff is in a first elongate rectangular configuration prior to implantation and a second curled cylindrical configuration when implanted around the site of the cancerous tumor;

wherein the cuff comprises two or more cuff electrodes disposed on the cuff, wherein each of the two or more cuff electrodes is activated independently; and

wherein the cuff expands in diameter by at least 50% without structural failure.

2. The lead of claim 1, further comprising an insulating flange disposed circumferentially on an exterior surface of the lead body, the insulating flange comprising an insulating material and configured to interrupt a direct electrical conduction path between a second zone and a different second zone, wherein the second zone and the different second zone function as electrodes.

3. The lead of claim 2, wherein the direct electrical conduction path is a straight-line electrical conduction path.

4. The lead of claim 1, wherein at a site of each second zone, the insulating material is disposed asymmetrically in thickness around a diameter of the lead body.

5. The lead of claim 1, wherein the thickness of the insulating material over the one or more second zones is less than 10 μm .

6. The lead of claim 1, wherein the thickness of the insulating material over the one or more first zones is less than 10 μm .

7. The lead of claim 1, wherein the insulating material over the one or more first zones is different than the insulating material over the one or more second zones.

8. The lead of claim 1, wherein the insulating material is selected from one or more of expanded polytetrafluoroethylene (ePTFE), polytetrafluoroethylene (PTFE), polyethylene-co-tetrafluoroethylene (ETFE), polyurethane, silicone, poly (p-xylylene) polymer, polyether block amide, nylon, iridium oxide, titanium oxide, and tantalum pentoxide.

9. The lead of claim 1, wherein the shape set material is nitinol.

10. The lead of claim 1, wherein the lead is configured to implement a switching mechanism to change which of the one or more cuff electrodes are in electrical communication with the lead.

11. A method of treating a cancerous tumor comprising: implanting a lead within a patient, the lead comprising a lead body having a proximal end and a distal end, the lead body comprising one or more conductors disposed within the lead body and providing electrical communication;

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an insulating material disposed circumferentially about and along the lead body over the one or more conductors, wherein the lead body comprises one or more first zones and one or more second zones, and wherein the insulating material over the one or more conductors in the first zones is thicker than the insulating material disposed over the one or more conductors in the second zones, each second zone comprising at least a portion of the one or more conductors and at least a portion of the insulating material to form an electric field generating electrode configured to generate one or more therapeutic electric fields at or near the site of a cancerous tumor; and

a solid insulating flange disposed continuously circumferentially on an exterior surface of the lead body, wherein the solid insulating flange comprises a radial distance greater than its width, and wherein the solid insulating flange has a first segment having a first radial distance and a second segment having a second radial distance, such that the insulating flange is not radially symmetric in cross-section;

generating one or more therapeutic electric fields at or near the site of a cancerous tumor from the one or more electric field generating electrodes; and

wherein the portion of the insulating material disposed about the one or more conductors in the second zones is configured to manipulate a strength of the one or more therapeutic electric fields generated by the electric field generating electrodes within the second zones; and

the second zones having a stronger electric field than the first zones.

12. The method of claim 11, wherein at least one second zone comprises a coiled electrode in electrical communication with at least one of the one or more conductors.

13. The method of claim 12, wherein the coiled electrode is implanted subcutaneously.

14. The method of claim 11, wherein at a site of each second zone, the insulating material is disposed asymmetrically in thickness around a diameter of the lead body.

15. The method of claim 14, the one or more therapeutic electric fields comprising applied electric field strengths selected from a range of between 1 V/cm to 10 V/cm.

16. The method of claim 11, the one or more therapeutic electric fields delivered at one or more frequencies selected from a range of between 100 kHz to 300 kHz.

17. The method of claim 11, wherein the one or more therapeutic electric fields are generated using currents ranging from 20 mAmp to 500 mAmp.

18. The method of claim 11, wherein the therapeutic electric field is delivered to a site of a cancerous tumor along more than one vector.

19. The method of claim 18, wherein the therapeutic electric field is delivered to a site of a cancerous tumor along at least two vectors, wherein the vectors are disposed at an angle with respect to one another of at least 10 degrees.

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